University of Engineering and Technology, Lahore

Name: Maria Abbas

Reg No: 2026(S) CYS_61

Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 1

Date: 16 Feb 2026

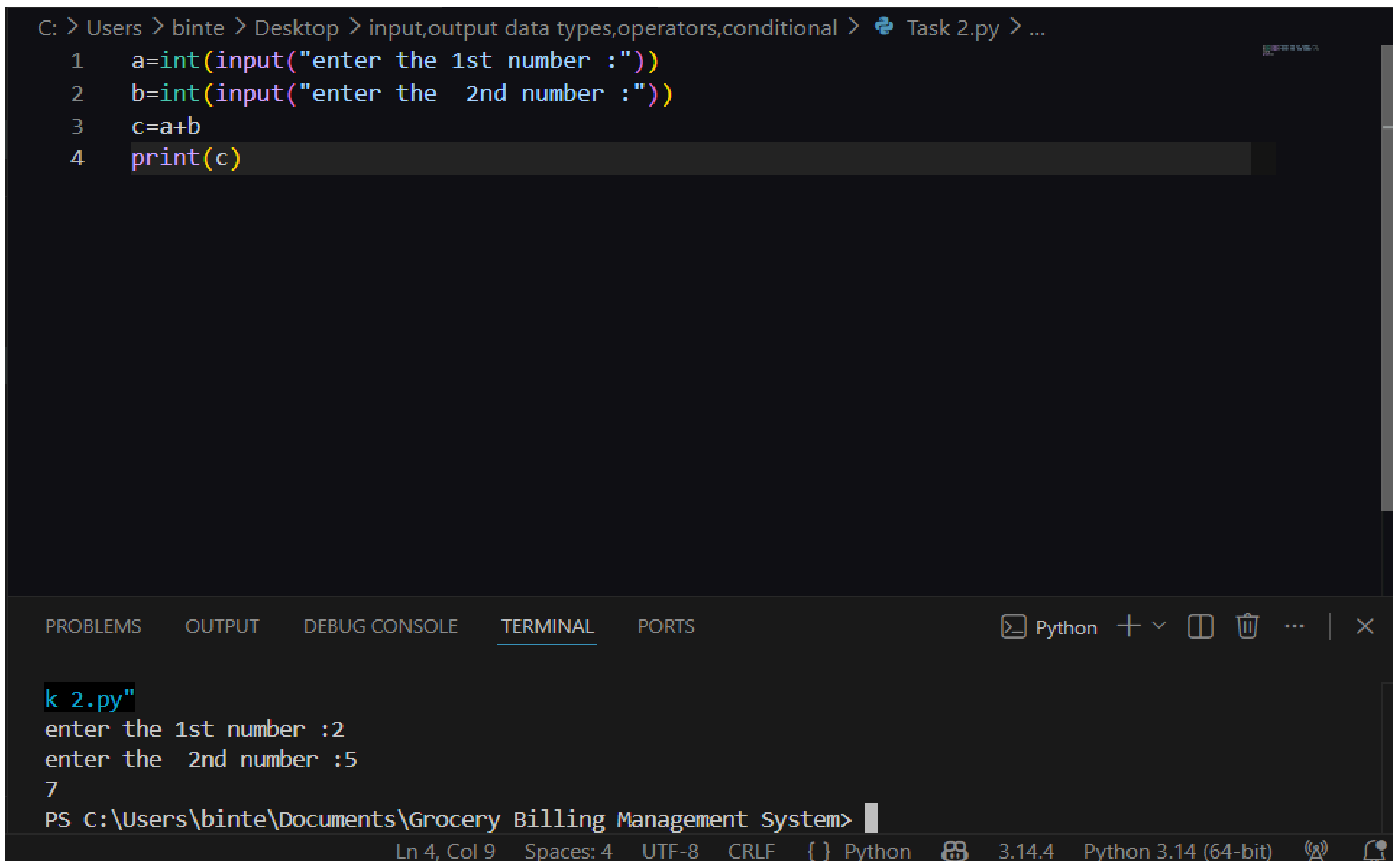
CLO1

Lab Tasks:

Input, Output data types, conditional statement, operators

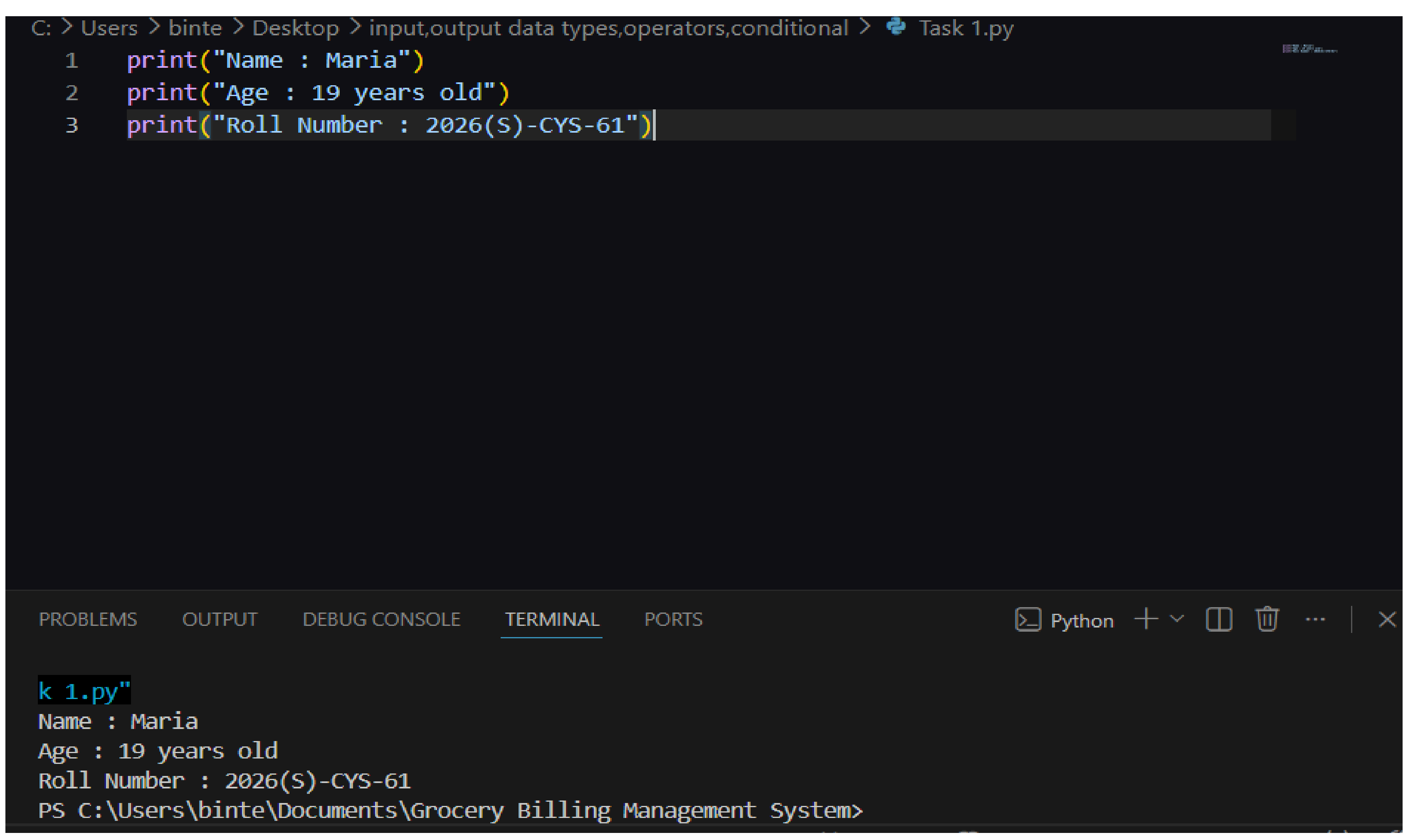
Task 1:

This programme takes input and gives sum of integers.



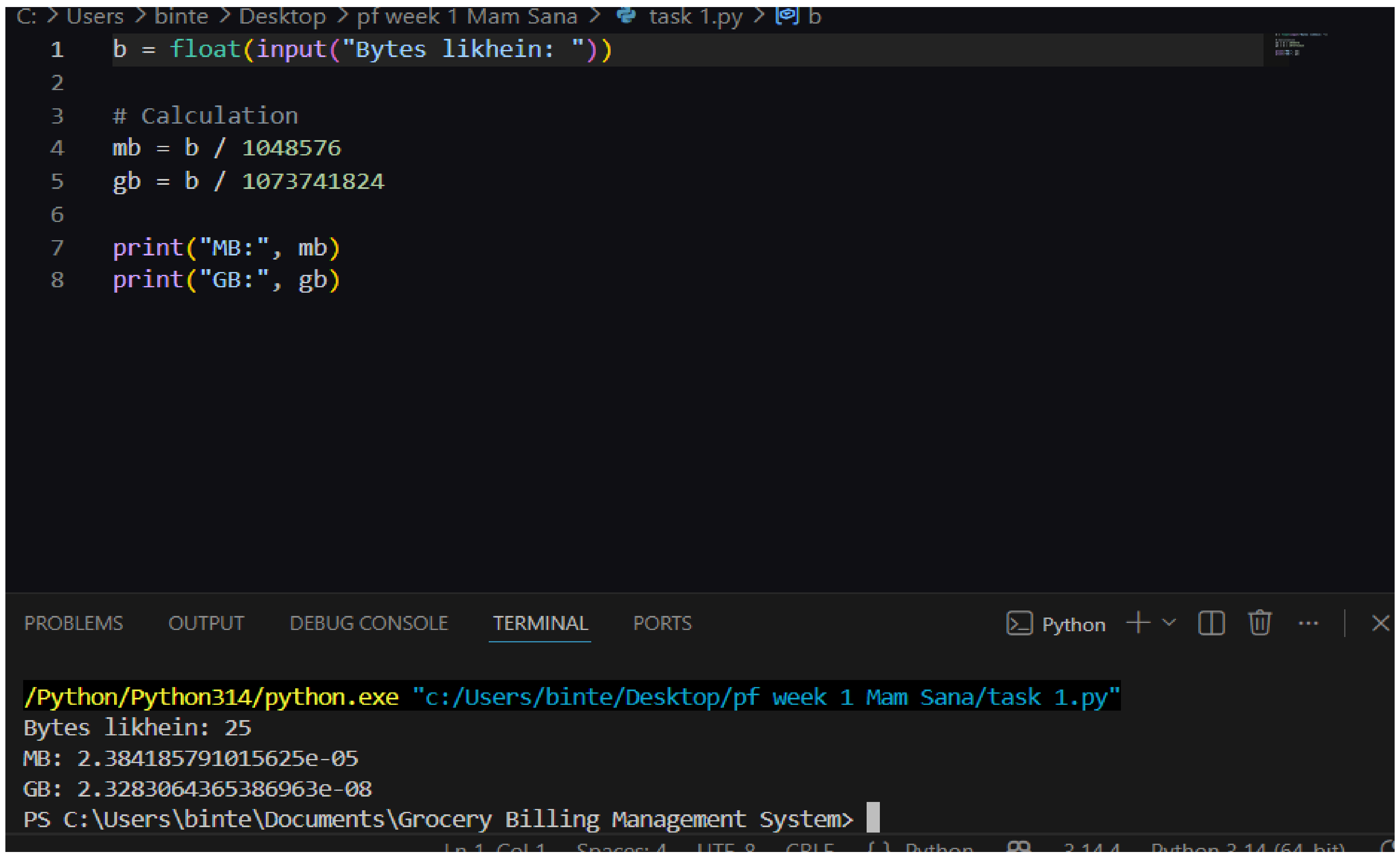
Task 2:

This programme prints input and gives us output.



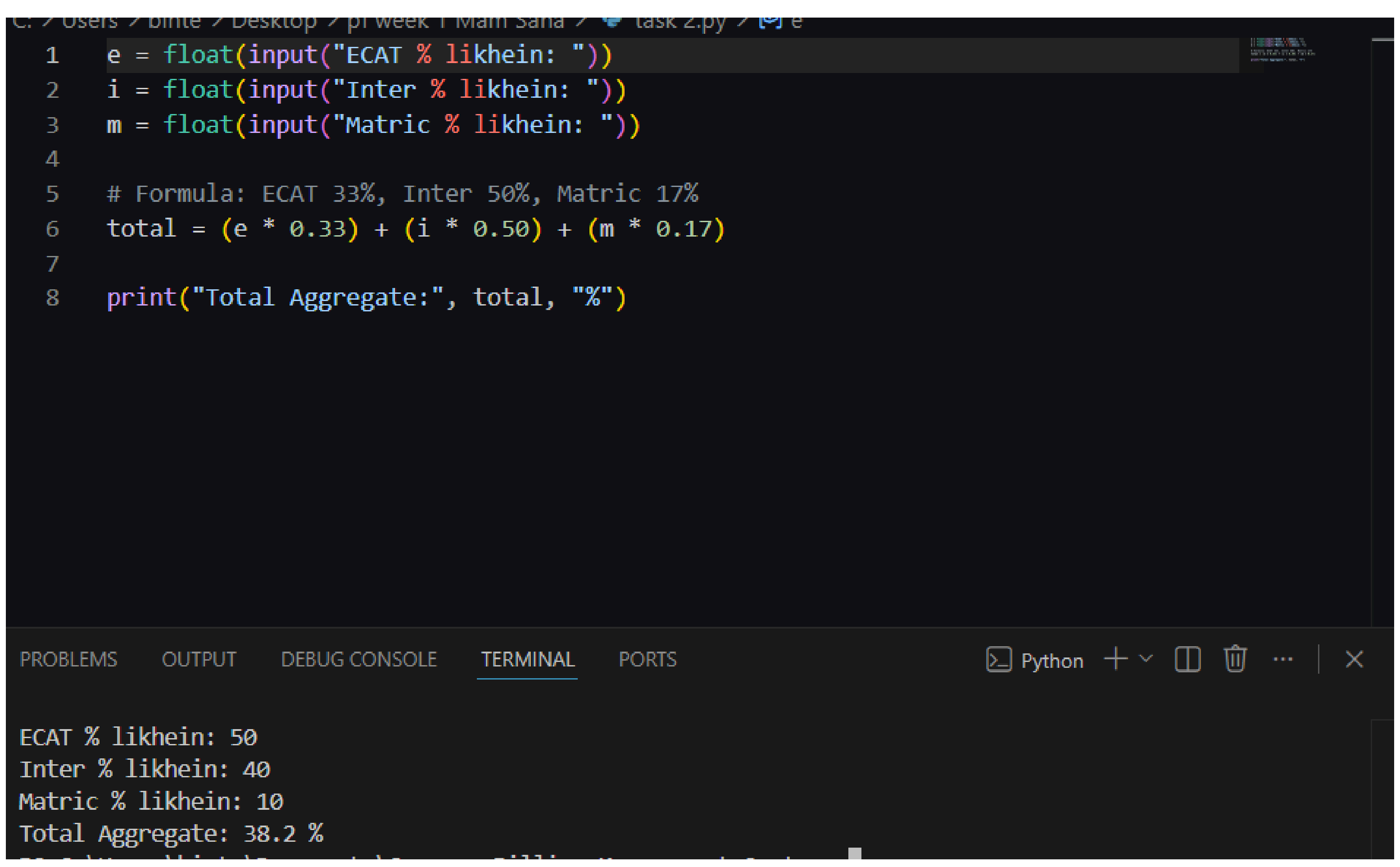
Task 3:

This programme takes a byte value from the user and converts it into Megabytes and Gigabytes.



Task 4:

This programme calculates the total university admission aggregate based on weighted percentages of ECAT, Inter, and Matric. It highlights the arithmetic operators to mathematical formula.



Task 5:

This programme calculates the geometric distance between two points and determines the specific quadrant of the first point.

```python
import math

x1 = float(input("x1: "))
y1 = float(input("y1: "))
x2 = float(input("x2: "))
y2 = float(input("y2: "))

# Distance Formula
d = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
print("Distance:", d)

# Point 1 ka Quadrant
if x1 > 0 and y1 > 0: print("Point 1: Quadrant 1")
if x1 < 0 and y1 > 0: print("Point 1: Quadrant 2")
if x1 < 0 and y1 < 0: print("Point 1: Quadrant 3")
if x1 > 0 and y1 < 0: print("Point 1: Quadrant 4")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
x2: 4
y2: 5
Distance: 2.8284271247461903
Point 1: Quadrant 1
Point 2: Quadrant 1
```

Task 6:

This programme solves a quadratic equation by calculating its discriminant to find the exact nature of its roots. It focuses on a conditional statement.

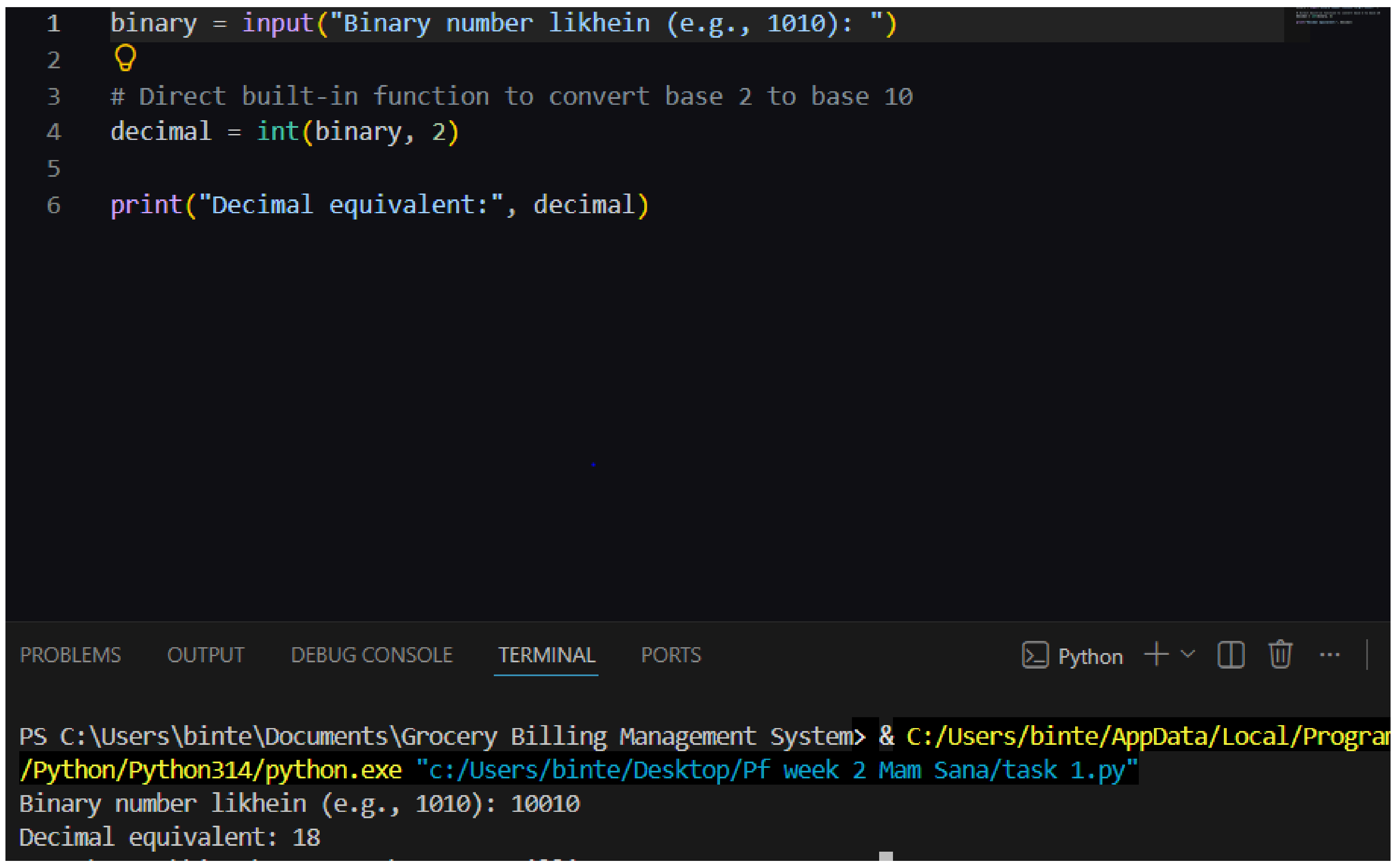
```python
import math

a = float(input("a ki value: "))
b = float(input("b ki value: "))
c = float(input("c ki value: "))

# Discriminant formula
disc = (b**2) - (4*a*c)
print("Discriminant:", disc)

if disc == 0:
    print("Roots are real, equal and rational.")
    root = -b / (2*a)
    print("Root:", root)
elif disc > 0:
    print("Roots are real, distinct.")
    root1 = (-b + math.sqrt(disc)) / (2*a)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
b ki value: 4
c ki value: 6
Discriminant: -32.0
Roots are imaginary.
```

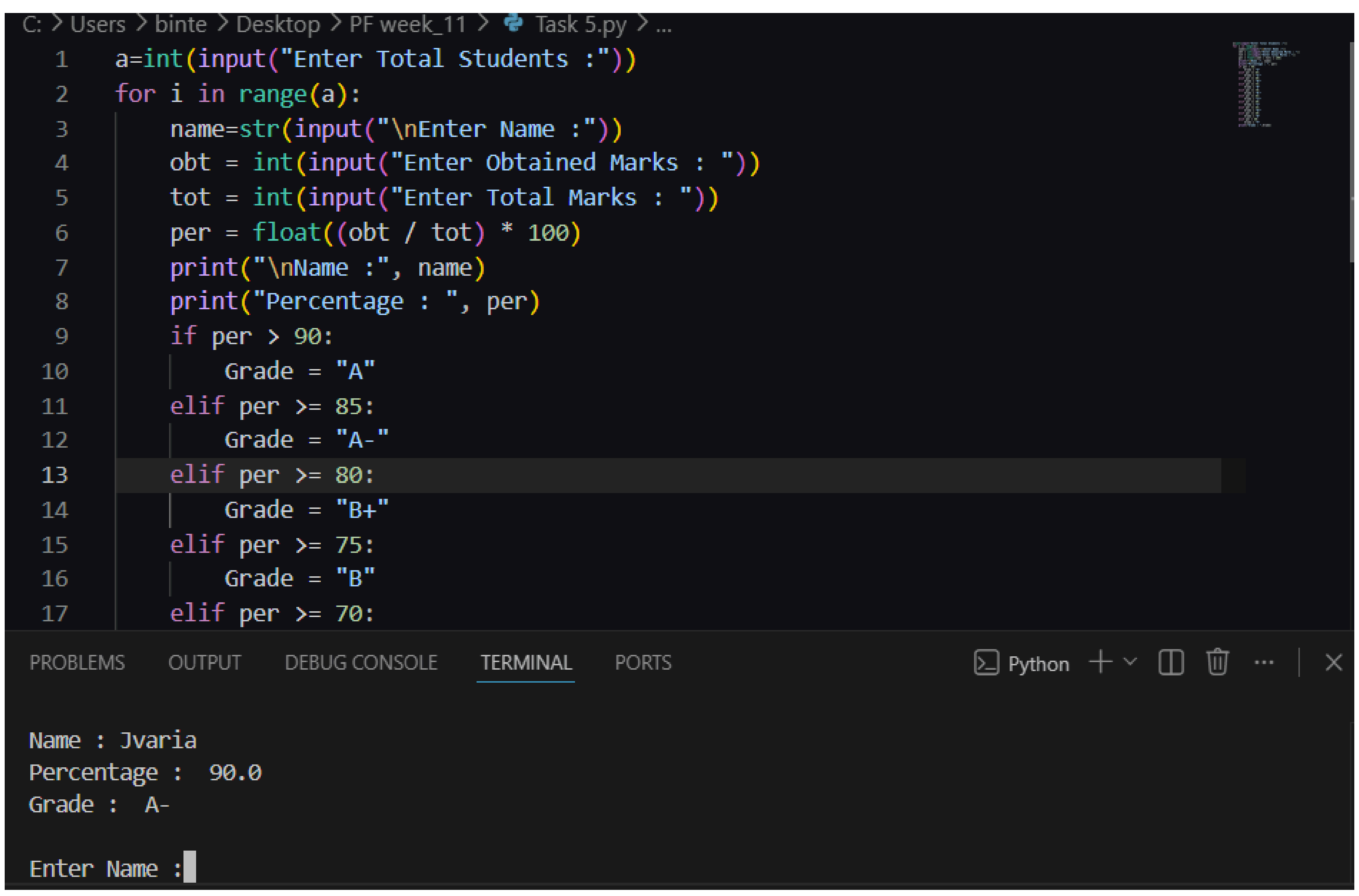
Task 7:

This programme converts a user binary number string into its standard decimal integer equivalent.



Task 8:

This programme calculates the grades of students.



Task 9:

This programme makes sum of two numbers.

```python
a=int(input("enter the 1st number :"))
b=int(input("enter the  2nd number :"))
c=a+b
print(c)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/Python/Python314/python.exe "c:/Users/binte/Desktop/PF week_11/Task 3.py"
enter the 1st number :20
enter the  2nd number :40
60
PS C:\Users\binte\Documents\Grocery Billing Management System>
```

Task 10:

This programme highlights how textual data is initialized and printed directly to screen without complexity.

```python
n = input("Enter Name :")
a = input("Enter Age :")
r = input("Enter Roll Number :")
print("Name : ", n)
print("Age : ", a)
print("Roll Number : ", r)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter Name :Name
Enter Age :19
Name :  Name
Roll Number :  61
```

Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 2

Date: 2 March 2026

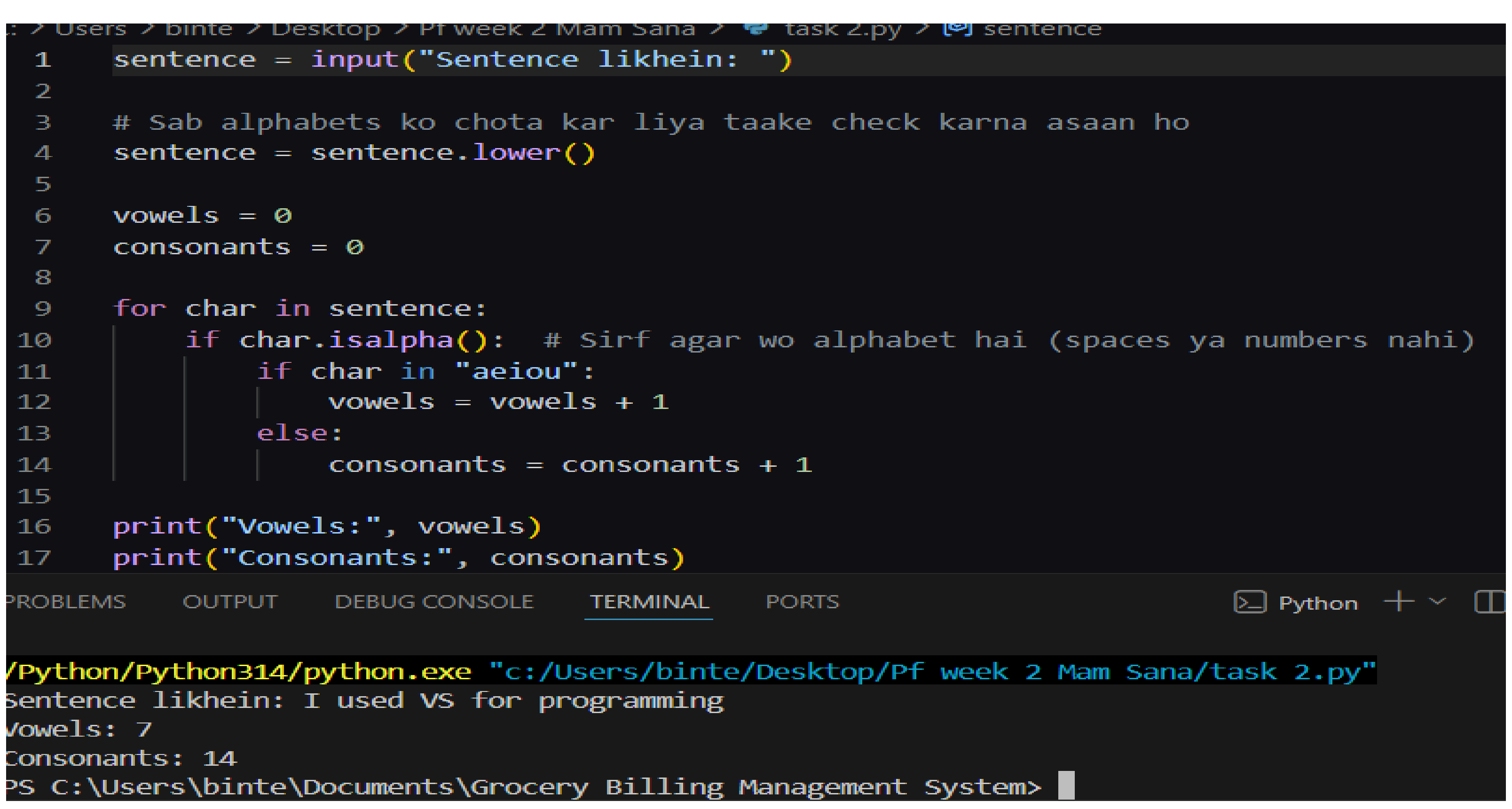
CLO1

Lab Tasks

For Loops:

Task 1:

This programme counts the total number of vowels and consonants in a user provided string by iterating each character using for loop.



```python
sentence = input("Sentence likhein: ")

# Sab alphabets ko chota kar liya taake check karna asaan ho
sentence = sentence.lower()

vowels = 0
consonants = 0

for char in sentence:
    if char.isalpha():  # Sirf agar wo alphabet hai (spaces ya numbers nahi)
        if char in "aeiou":
            vowels = vowels + 1
        else:
            consonants = consonants + 1

print("Vowels:", vowels)
print("Consonants:", consonants)
```

```
/Python/Python314/python.exe "c:/Users/binte/Desktop/Pf week 2 Mam Sana/task 2.py"
Sentence likhein: I used VS for programming
Vowels: 7
Consonants: 14
PS C:\Users\binte\Documents\Grocery Billing Management System>
```

Task 2:

This programme generates a secure password based on custom user preference for length and character type.

```python
import random

length = int(int(input("Password length kitni chahiye? ")))
use_upper = input("Uppercase shamil karein? (y/n): ").lower()
use_digits = input("Digits shamil karein? (y/n): ").lower()
use_special = input("Special characters shamil karein? (y/n): ").lower()

# Default lowercase har haal mein hoga
pool = "abcdefghijklmnopqrstuvwxyz"

if use_upper == 'y':
    pool = pool + "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
if use_digits == 'y':
    pool = pool + "0123456789"
if use_special == 'y':
    pool = pool + "!@#$%^&*()"

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
Uppercase shamil karein? (y/n): y
Digits shamil karein? (y/n): y
Special characters shamil karein? (y/n): y
Generated Password: h$Cm7c
PS C:\Users\binte\Documents\Grocery Billing Management System>
```

Task 3:

This programme loops through a sequence of numbers from 1 to 5 and print each integers.

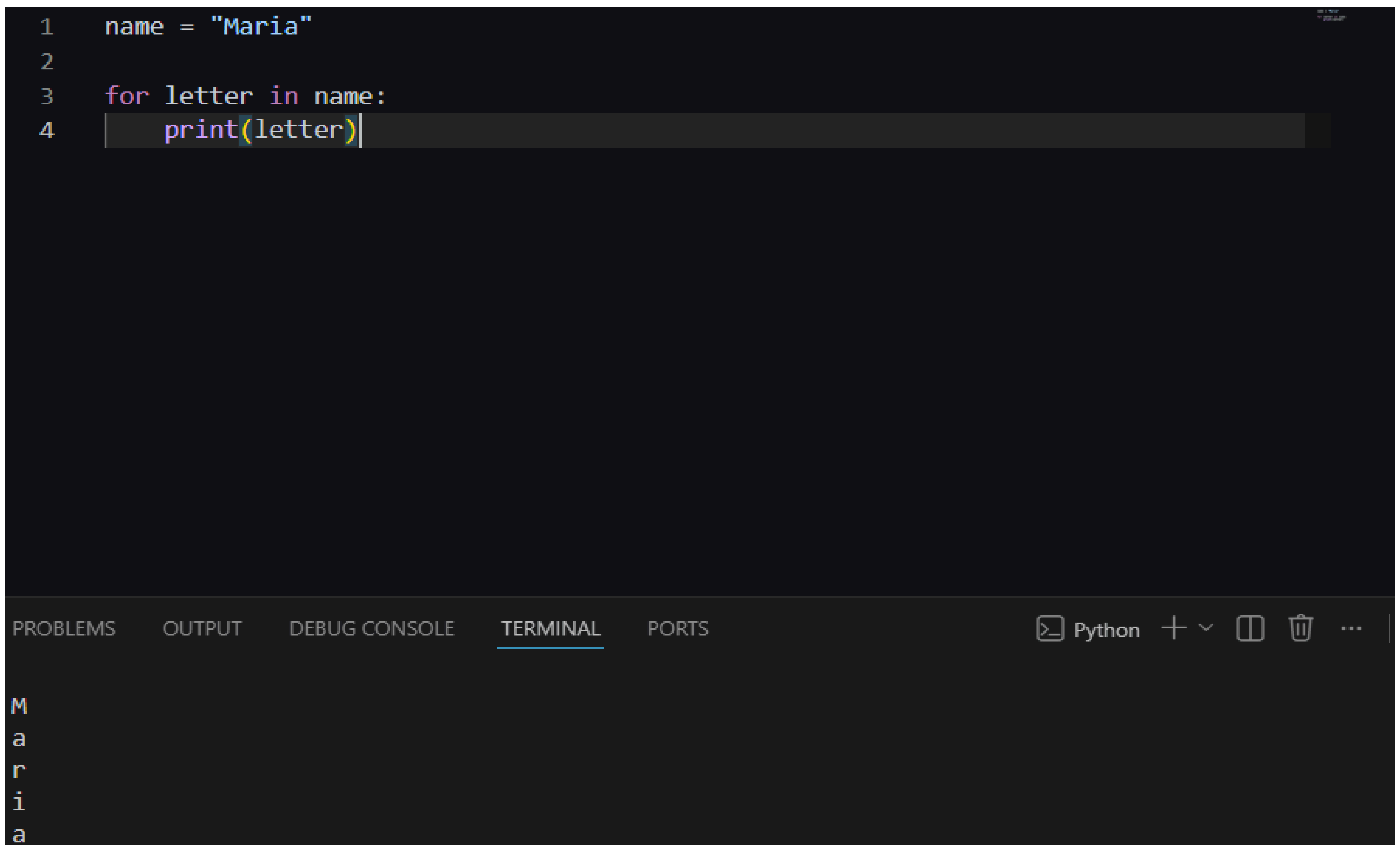
```python
for i in range(1, 6):
    print(i)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
2
3
4
5
PS C:\Users\binte\Documents\Grocery Billing Management System>
```

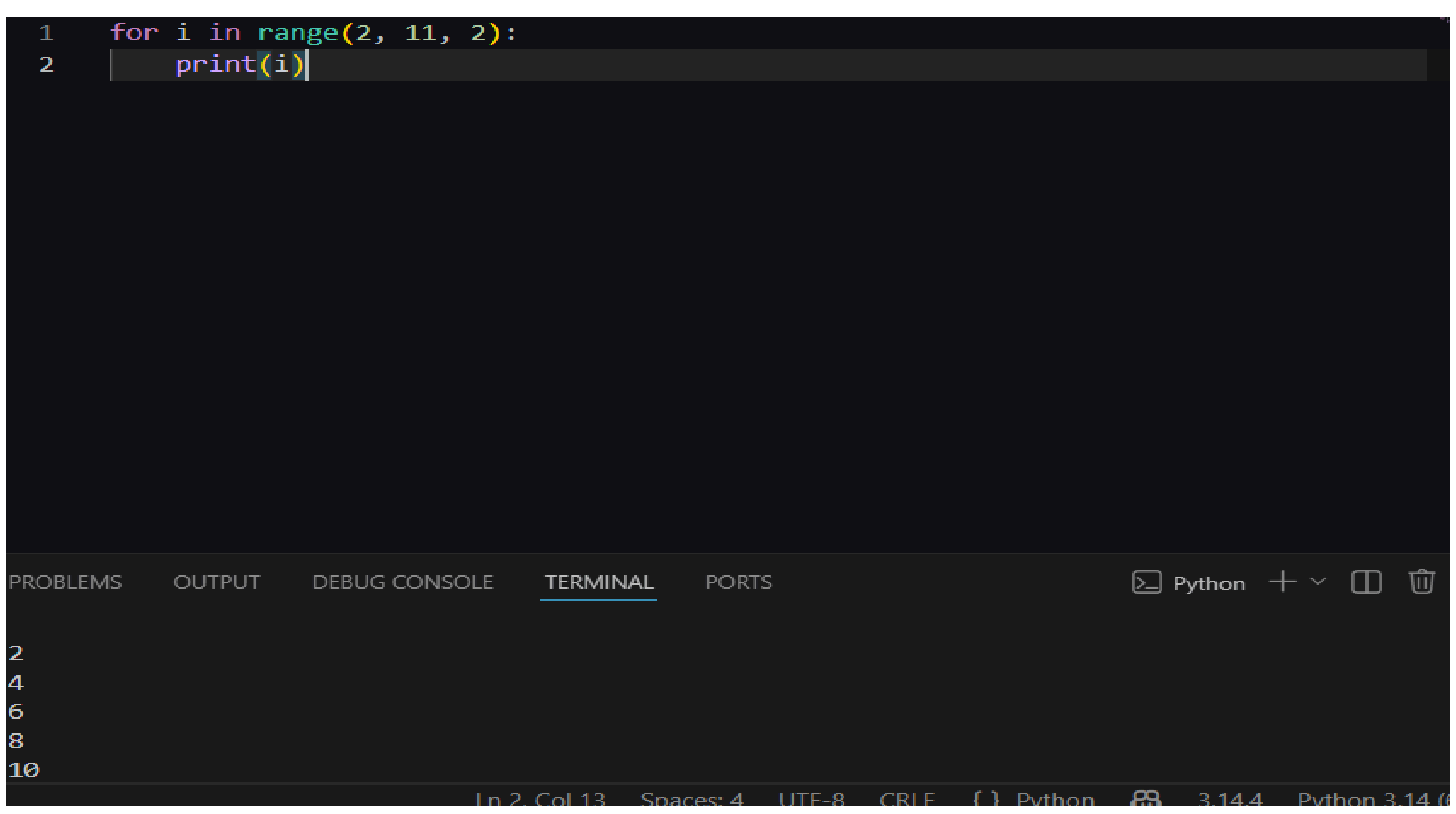
Task 4:

This programme loops through a string variable character by character to print each letter on new line.



Task 5:

This programme prints even numbers from 2 to 10 by using a custom step value inside a loop.



Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 3

Date: 16 March 2026

CLO1

Lab Tasks

While loops:

Task 1

This programme uses a while loop to print star pyramid. The outer loop controls the rows and inner loop controls spaces.

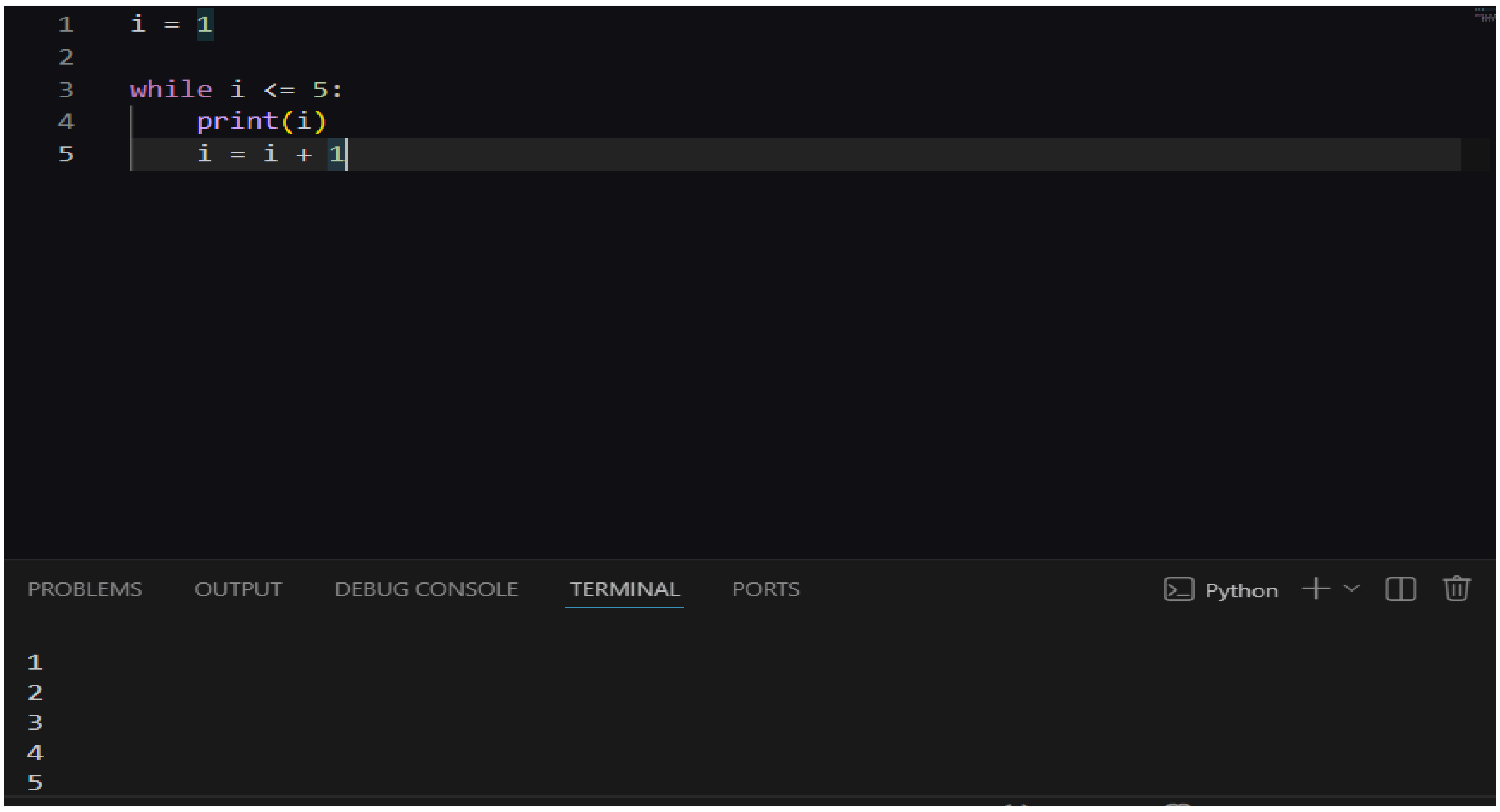
```python
i=1
while(i<=4):
    j=0
    while(j<4-i):
        print(" ", end=" ")
        j=j+1
    j=0
    while j<(i*2-1):
        print("*", end=" ")
        j=j+1
    print()
    i=i+1
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
      *
    * * *
  * * * * *
* * * * * * *
```

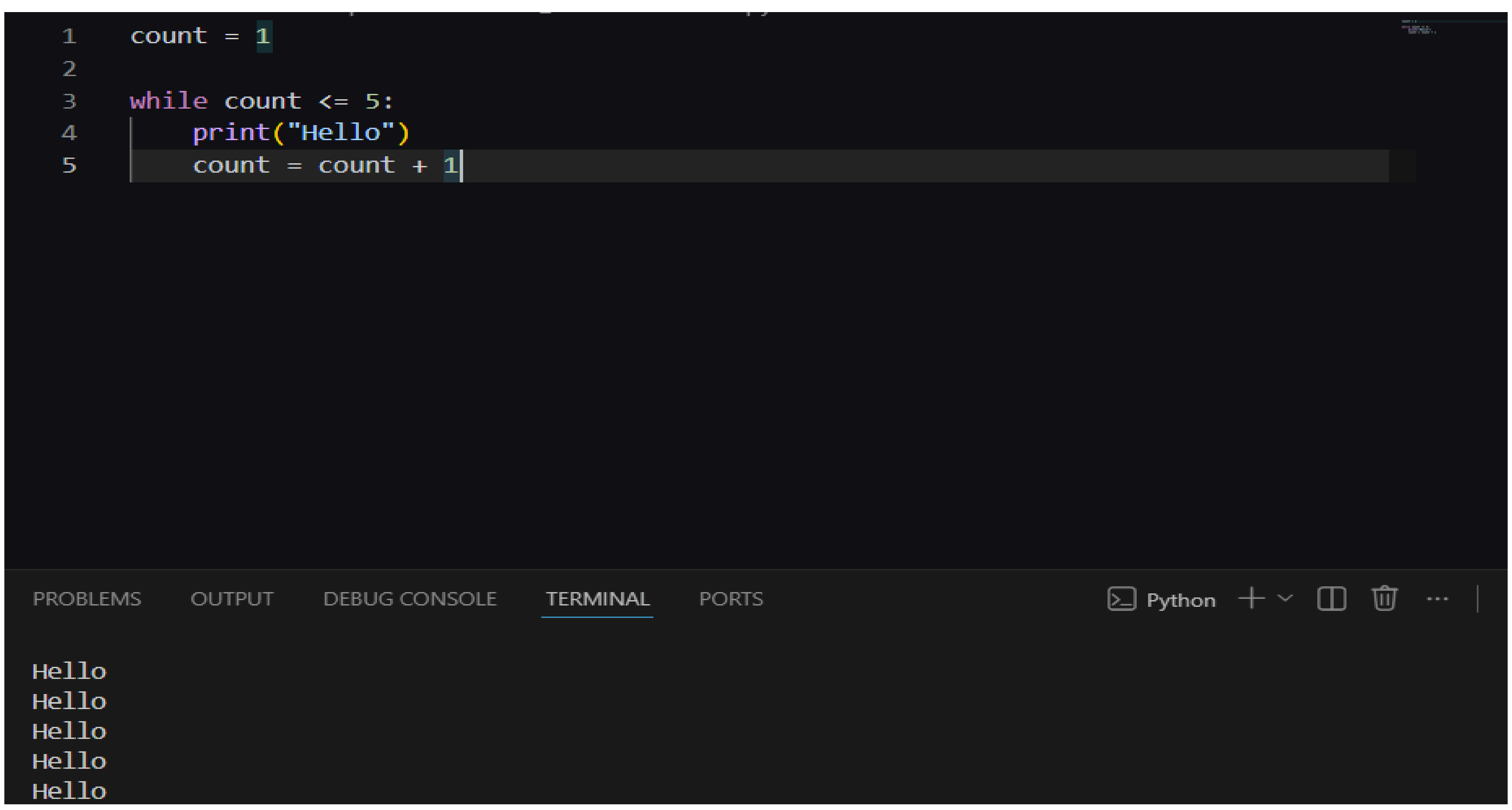
Task 2:

This programme uses a while loop to count and print numbers from 1 to 5. It repeats until a condition become false.



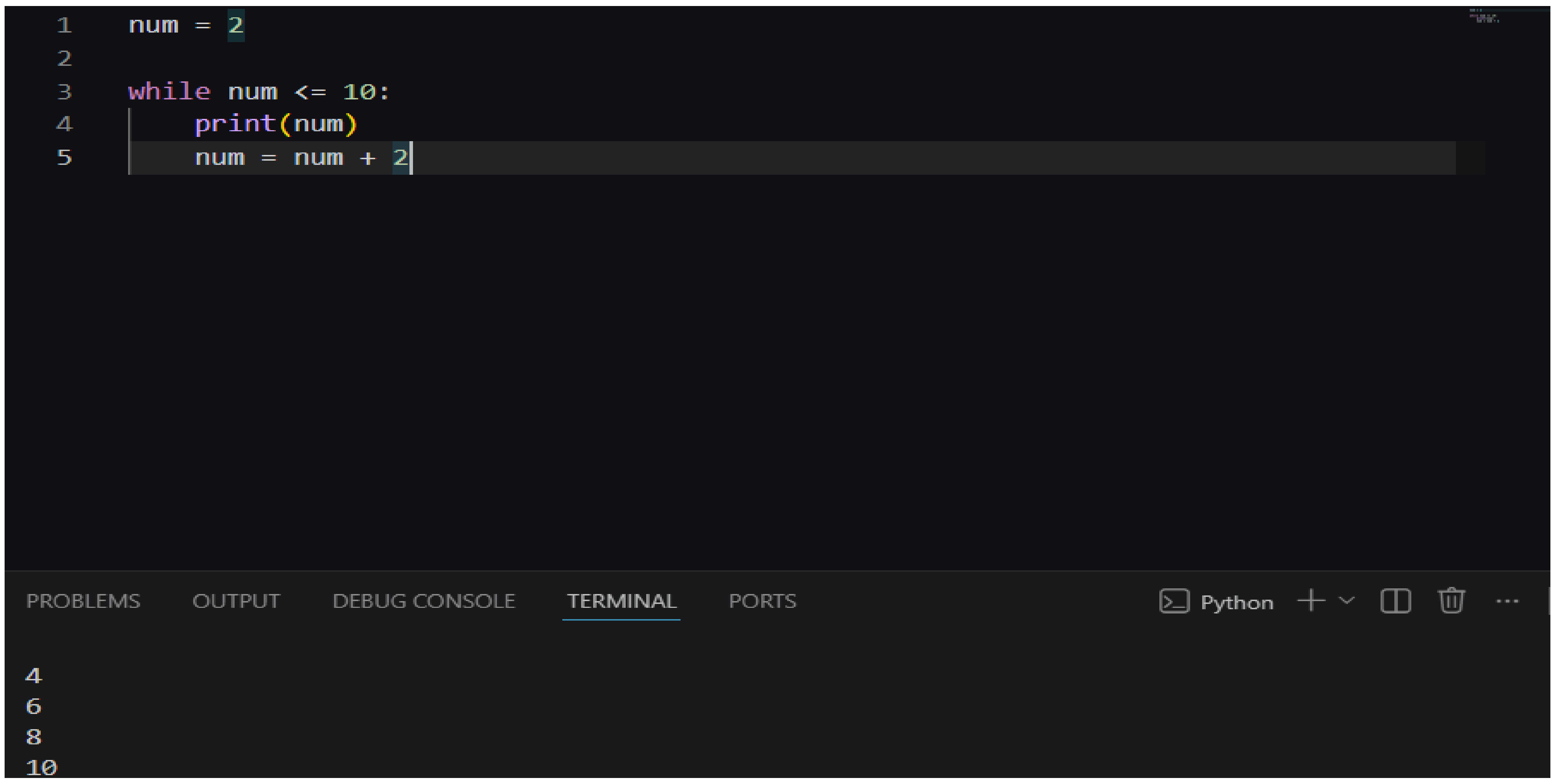
Task 3:

This programme uses a while loop to print the word "Hello" five times on the screen. It repeat a specific task using a counter variable.



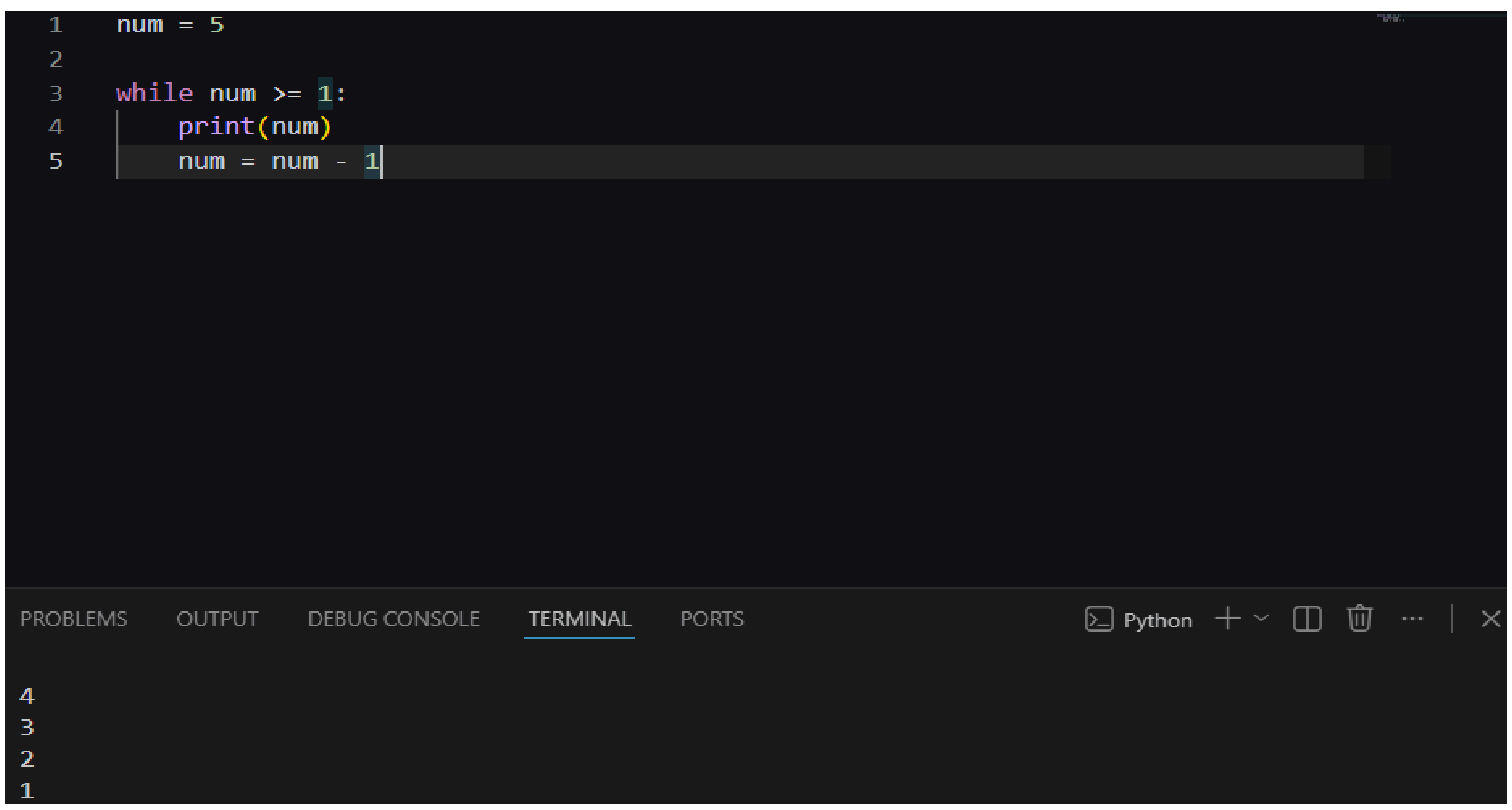
Task 4:

This programme uses a while loop to skip count and print even numbers up to 10.



Task 5:

This programme uses a while loop to print a reverse countdown from 5 down to 1.



Task 6:

This *programme uses a while loop to print a star (*) pattern. It prints 2 stars in the first line, 5 stars in the second line, and 7 stars in the third line.*

```
C: > Users > binte > Desktop > PF Lab week_12 > 1 while.py > ...
rows = 3
i = 1

while i <= rows:
    if i == 1:
        print("*" * 2)
    elif i == 2:
        print("*" * 5)
    else:
        print("*" * 7)

    i = i + 1
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                Python

/Python/Python314/python.exe "c:/Users/binte/Desktop/PF Lab week_12/1 while.py"
**
*****
*******
```

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Section: B

Assignment: PF Lab week 4

Date: 23 March 2026

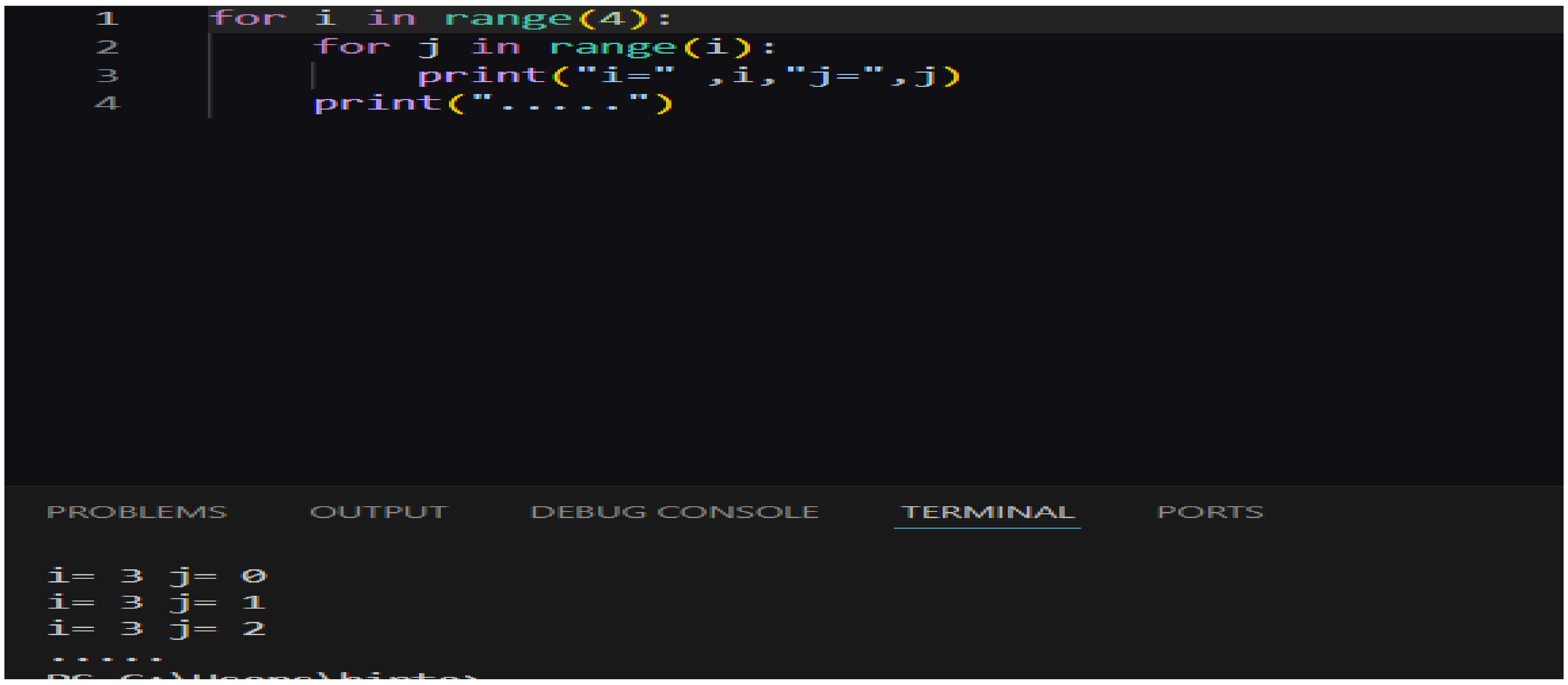
CLO1

Lab Tasks

Nested loops

Task 1

This programme uses a nested for loops to track and print value of variables (i) and (j) together



Task 2:

This programme uses nested for loop to print pair of numbers like coordinates. It highlights how inner loop repeats inside the outer loop to generate structure pattern.

Task 3:

This programme use nested for loop to print a growing pattern numbers on screen.

```python
for i in range(4):
    for j in range(1,1+i):
        print(j,end="")
    print()
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
1
12
123
```

Task 4:

This programme creates a centered star pyramid. The outer loop controls the rows while inner loop manage spacing.

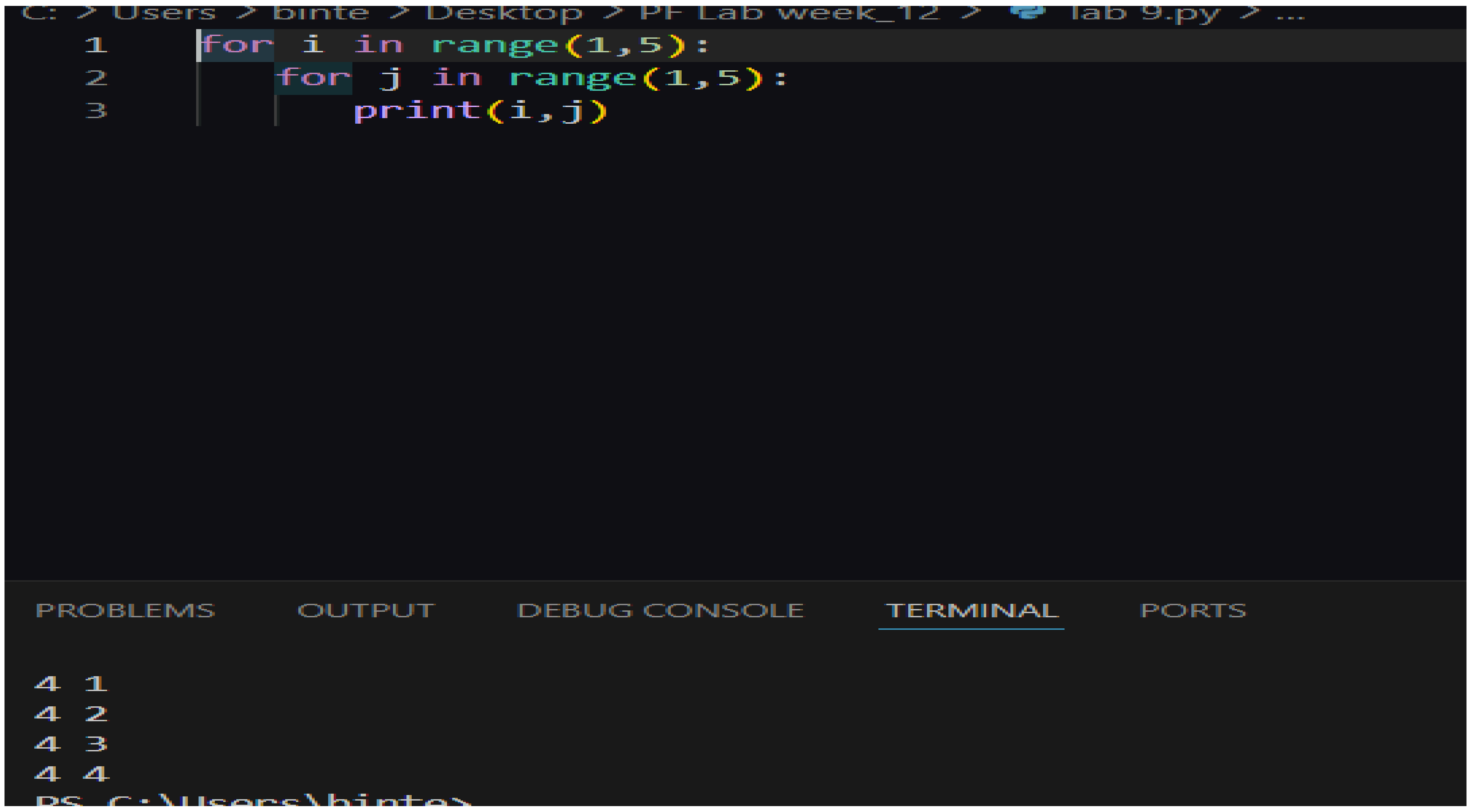
```python
i=1
while(i<=4):
    j=0
    while(j<4-i):
        print(" ", end=" ")
        j=j+1
    j=0
    while j<(i*2-1):
        print("*", end=" ")
        j=j+1
    print()
    i=i+1
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
      *
    * * *
  * * * * *
* * * * * * *
PS C:\Users\binte>
```

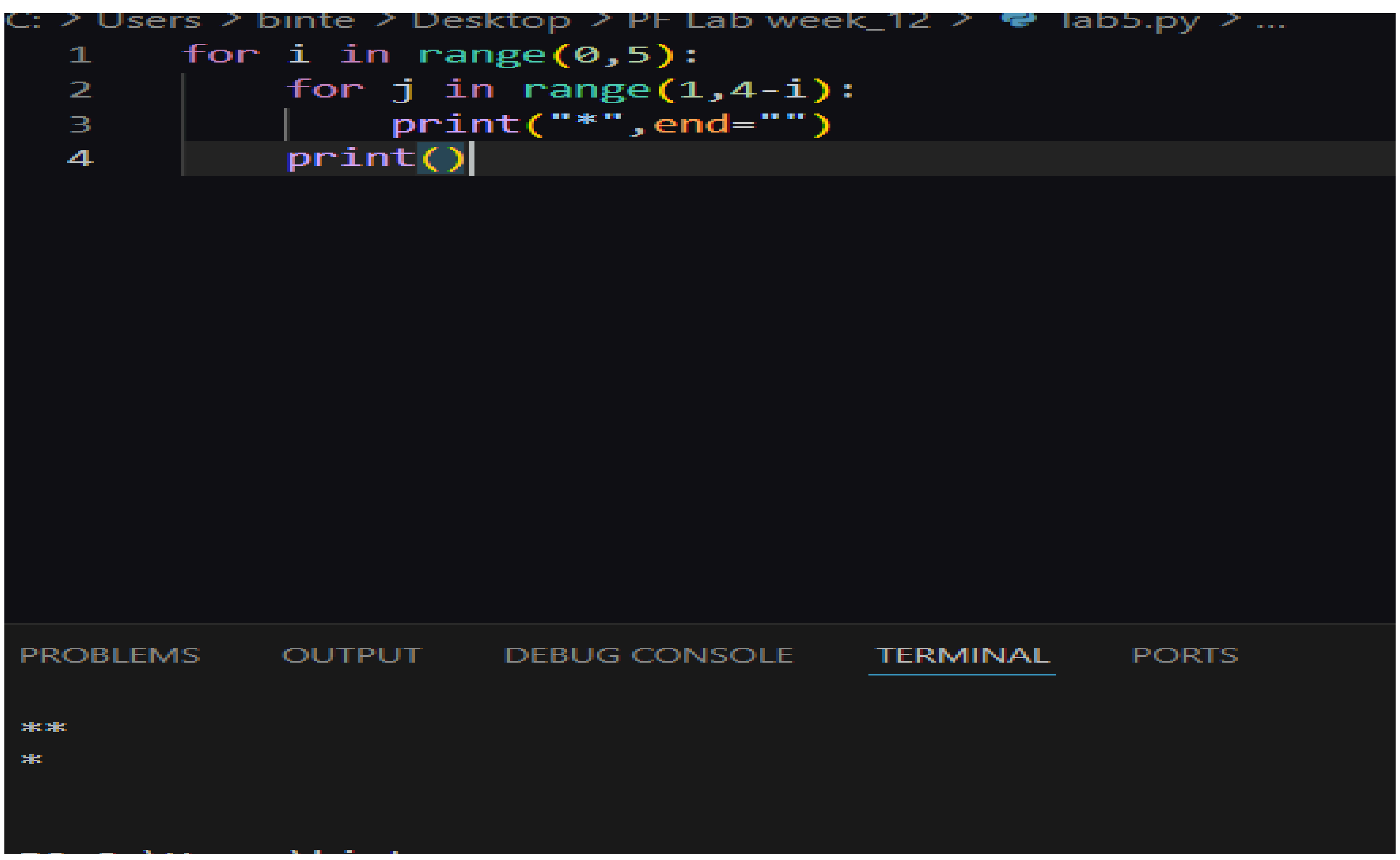
Task 5:

This programme uses nested loop to generate and print combination of numbers between 1 and 4.It demonstrates how an inner loop runs its full cycle for every single iteration of outer loop.



Task 6:

This programme uses nested loop to print fewer stars.



Task 7:

This programme uses nested loop to create simple right-side angle star triangle. The outer loop handles the row count while inner loop prints increasing number of stars row by row.

Task 8:

This program uses nested *for loops* combined with an if condition to print a growing triangle of repeating numbers. The inner loop checks the row value to print the current line's number multiple times.

```python
for i in range(1, 6):
    for j in range(1, 6):
        if j <= i:
            print(i, end=" ")
    print()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

Task 9:

This program uses two separate sets of nested *for loops* to create a sideways diamond star pattern. The first part handles an increasing star pattern, while the second part handles a decreasing star pattern.

```python
# Pehla hissa: Stars barh rahe hain (1 se 4)
for i in range(1, 5):
    for j in range(i):
        print("*", end=" ")
    print()  # Line change karne ke liye

# Doosra hissa: Stars kam ho rahe hain (3 se 1
for i in range(3, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()  # Line change karne ke liye
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* * * *
* * *
* *
*
```

Task 10:

This program prints consecutive numbers in a triangle pattern using nested loops.

The variable num increases by 1 after every number is printed.

```python
num = 1

for i in range(1, 6):
    for j in range(i):
        print(num, end=" ")
        num = num + 1
    print()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
2 3
4 5 6
7 8 9 10
11 12 13 14 15
PS C:\Users\binte>
```

University of Engineering and Technology, Lahore

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Degree: Cyber security

Section: B

Assignment: PF Lab week 5

Date: 30 March 2026

CLO2

Lab Tasks

Functions

Task 1:

This program uses Python's built-in len() function to calculate the total number of characters in a string. It shows how a function takes an input value and returns its size to be stored and printed.

```python
a=len("Maria")
print(a)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte> & C:/Users/binte/AppData/Local/Programs/Python/Python314/
/Desktop/PF Lab week_12/lab 7.py"
5
PS C:\Users\binte>
```

Task 2:

This program uses Python's built-in max() function to find the largest number from a given group of values. It compares the numbers and returns the highest value to be saved and printed.

```python
a=max(12,15,78)
print(a)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
/Desktop/PF Lab week_12/lab 6.py"
78
PS C:\Users\binte>
```

Task 3:

This program uses a custom user-defined function called fact() to calculate the factorial of a number. It then reuses this function to compute and display mathematical permutation and combination values based on user input.

```python
def fact(num):
    f = 1
    for i in range(1, num + 1):
        f = f * i
    return f

n = int(input("enter n value: "))
r = int(input("enter r value: "))

# Formulas
permutation = fact(n) / fact(n - r)
combination = fact(n) / (fact(r) * fact(n - r))

print("Permutation:", permutation)
print("Combination:", combination)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
enter n value: 2
enter r value: 4
Permutation: 2.0
Combination: 0.08333333333333333
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 4:

This program uses a lambda function to convert text to uppercase and a user-defined function to reverse it. It demonstrates how to modify and flip an input string step by step.

```python
# Uppercase karne ke liye lambda
to_upper = lambda s: s.upper()

# String ko ulta karne ka basic UDF
def invert(text):
    # Python mein [::-1] string ko direct ulta kar deta hai
    print("Ulta string:", text[::-1])

user_string = input("String likhein: ")

uppercase_string = to_upper(user_string)
invert(uppercase_string)
```

```
ROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

S C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python3
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/task 3.py"
tring likhein: hello
lta string: OLLEH
S C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 5:

This program uses two user-defined functions to convert temperatures between Celsius and Fahrenheit. It shows how to pass user input into mathematical formulas to calculate and return the converted results.

```python
def to_celsius(f):
    return (f - 32) * 5 / 9

def to_fahrenheit(c):
    return (c * 9 / 5) + 32

# Dono convertors ko test karne ke liye simple inputs
fah = float(input("Fahrenheit temperature : "))
print("Celsius mein:", to_celsius(fah))

cel = float(input("Celsius temperature : "))
print("Fahrenheit mein:", to_fahrenheit(cel))
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Fahrenheit temperature : 90
Celsius mein: 32.22222222222222
Celsius temperature : 30
Fahrenheit mein: 86.0
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 6:

This program uses a user-defined function and a for loop to calculate a student's total GPA for the semester. It loops through each subject to take grade points and credit hours, then uses the GPA formula to print the final result.

```python
def calculate_gpa():
    subjects = int(input("subjects in semester? "))

    total_grade_points = 0
    total_credits = 0

    for i in range(1, subjects + 1):
        print(f"\nSubject {i}:")
        gp = float(input("Grade Point (GP): "))
        ch = int(input("Credit Hours (CH): "))

        # Calculation formula ke mutabiq
        total_grade_points = total_grade_points + (gp * ch)
        total_credits = total_credits + ch

    gpa = total_grade_points / total_credits
    print("\n--- Final Result ---")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
Credit Hours (CH): 4

--- Final Result ---
 Semester GPA : 55.71
```

Task 7:

This program uses a simple user-defined function called add() to calculate the sum of two numbers. It takes two integers as input from the user, passes them into the function, and prints the total sum.

```python
def add(a, b):
    return a + b

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

result = add(x, y)
print("Sum =", result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Enter first number: 10
Enter second number: 20
Sum = 30
```

Task 8:

This program uses a user-defined function called square() to calculate the mathematical square of a number. It takes an integer input from the user, multiplies it by itself, and returns the result to be printed.

```python
def square(num):
    return num * num

n = int(input("Enter a number: "))

print("Square =", square(n))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Enter a number: 32
Square = 1024
```

Task 9:

This program uses a user-defined function combined with an if-else condition to check whether a number is even or odd. It uses the modulus operator (%) to determine if the number is perfectly divisible by 2.

```python
def check(num):
    if num % 2 == 0:
        print("Even")
    else:
        print("Odd")

n = int(input("Enter a number: "))
check(n)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Enter a number: 7
Odd
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

University of Engineering and Technology, Lahore

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Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 6

Date: 20 April 2026

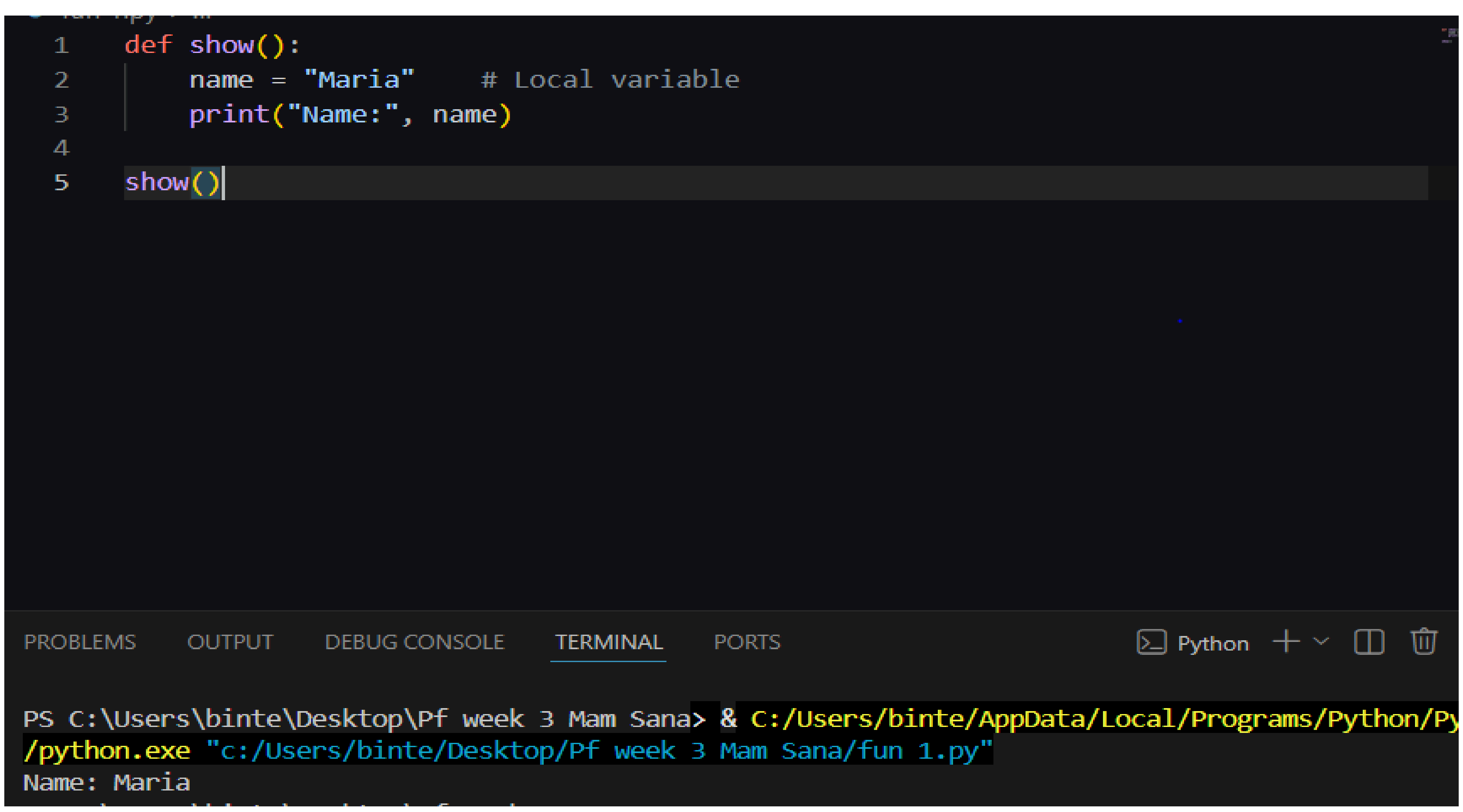
CLO2

Lab Tasks

Local and Global variables

Task 1:

Local variable



Global variable

```python
name = "Maria"     # Global variable

def show():
    print("Name:", name)

show()
print("Name:", name)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Py
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Name: Maria
Name: Maria
```

Task 2:

Local variables

```python
def add():
    a = 10
    b = 20
    print("Sum =", a + b)

add()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pyt
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Sum = 30
```

Global variables

```python
message = "Welcome"

def show():
    print(message)

def display():
    print(message)

show()
display()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/P
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Welcome
Welcome
```

Task 3:

Local variables and global variables

```python
x = 100        # Global variable

def show():
    x = 50    # Local variable
    print("Inside Function:", x)

show()
print("Outside Function:", x)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/L
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Inside Function: 50
Outside Function: 100
```

University of Engineering and Technology, Lahore

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Section: B

Assignment: PF Lab week 7

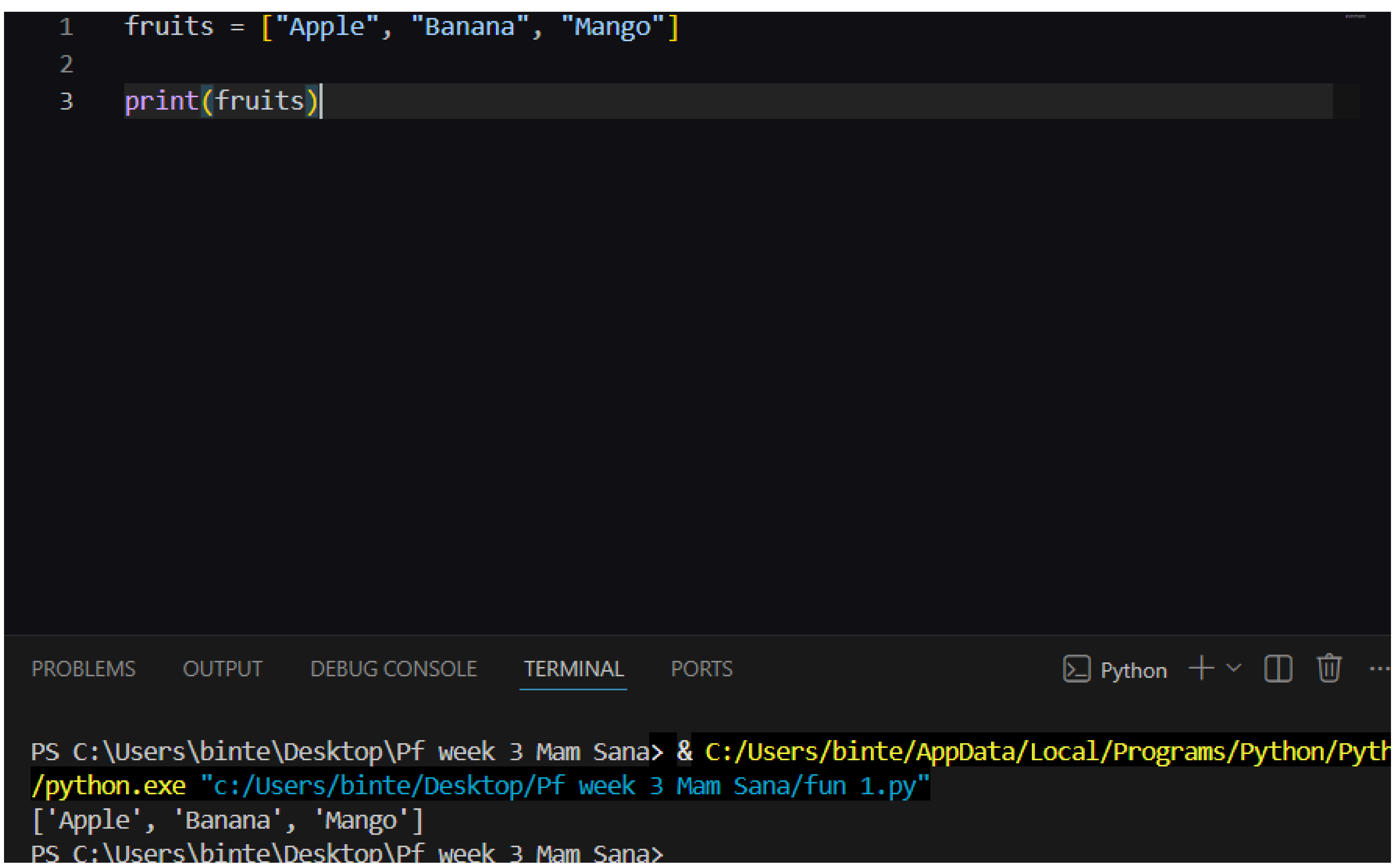
Date: 27 April 2026

CLO3

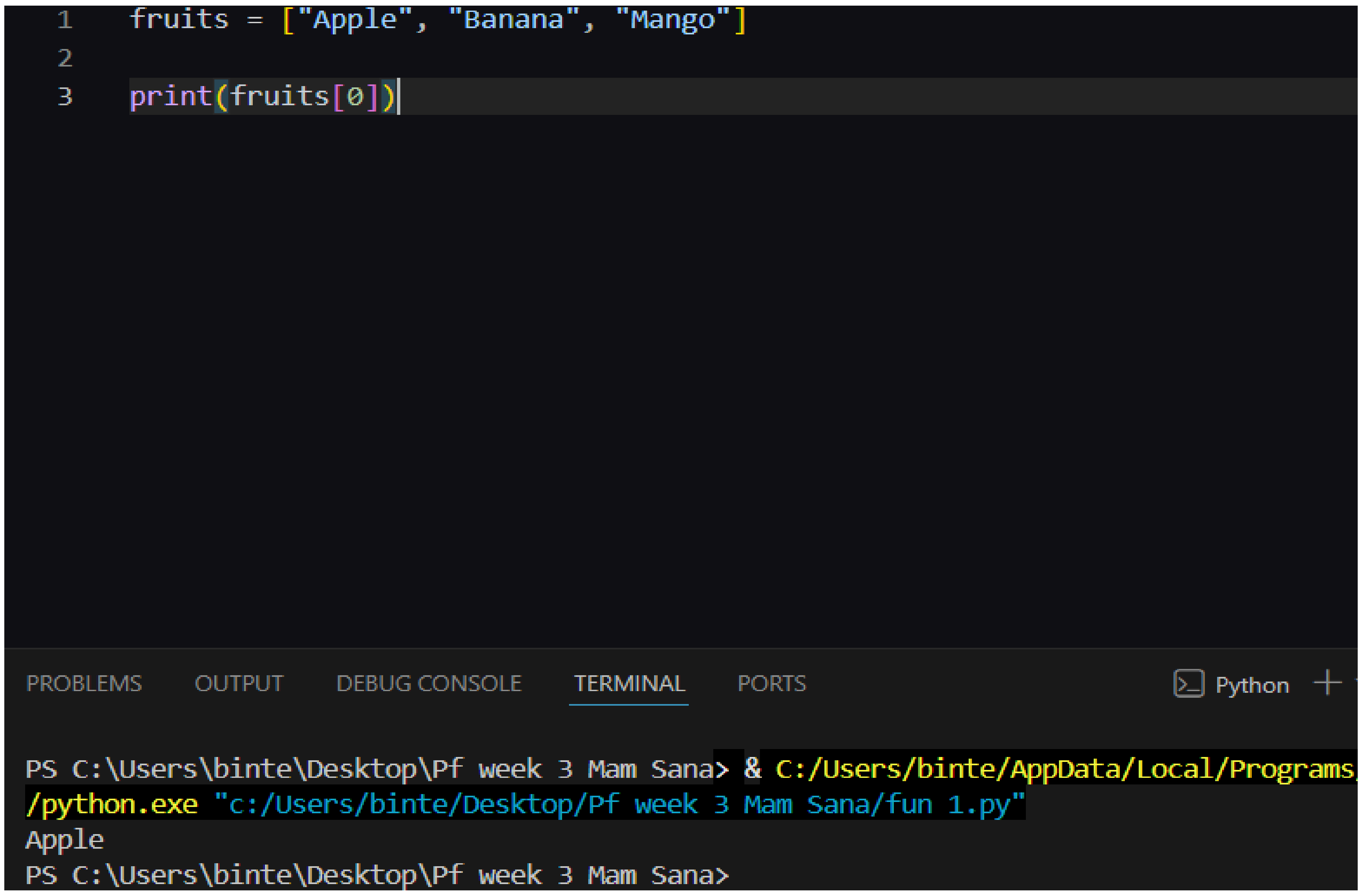
Lab Tasks

List

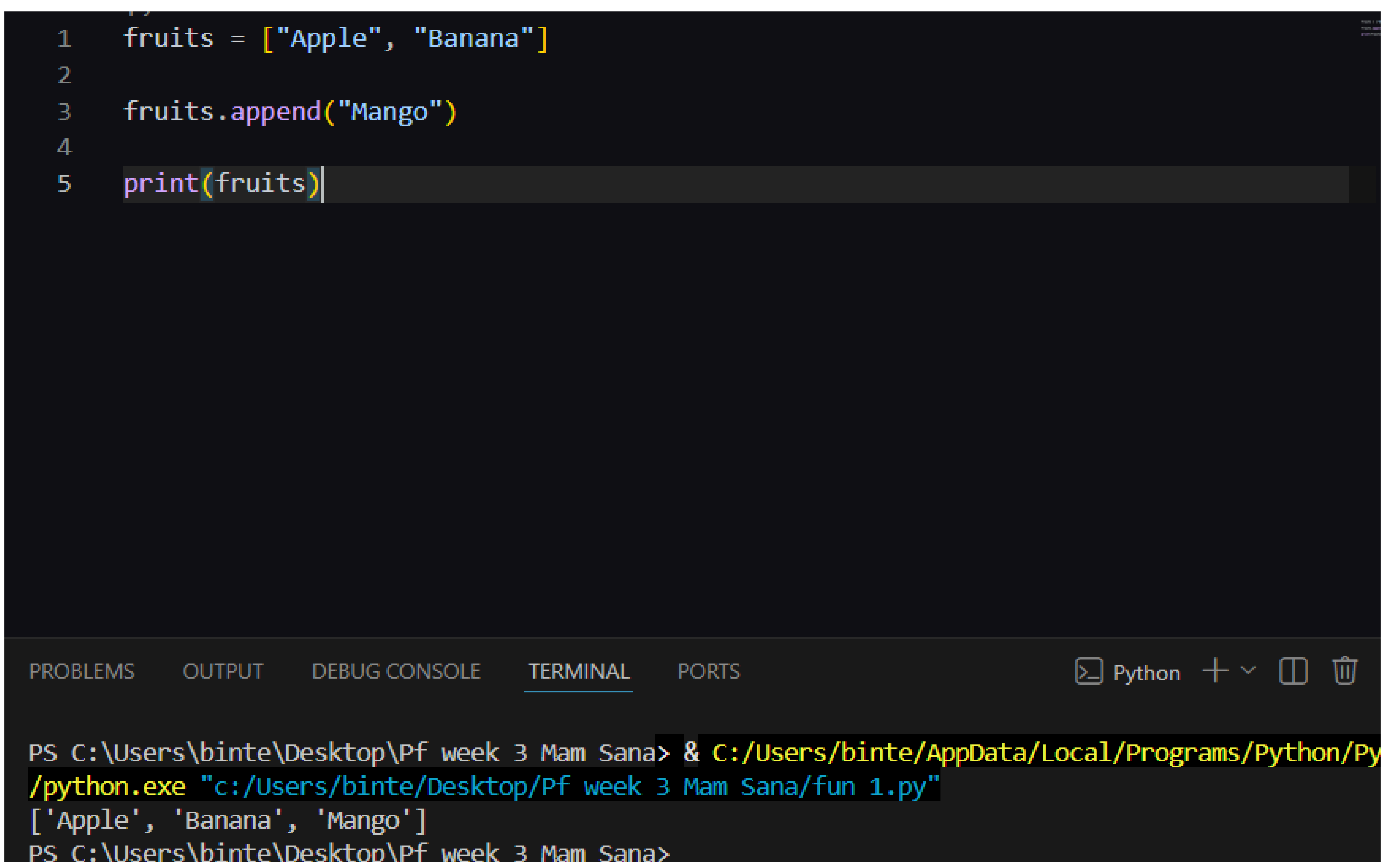
Task 1



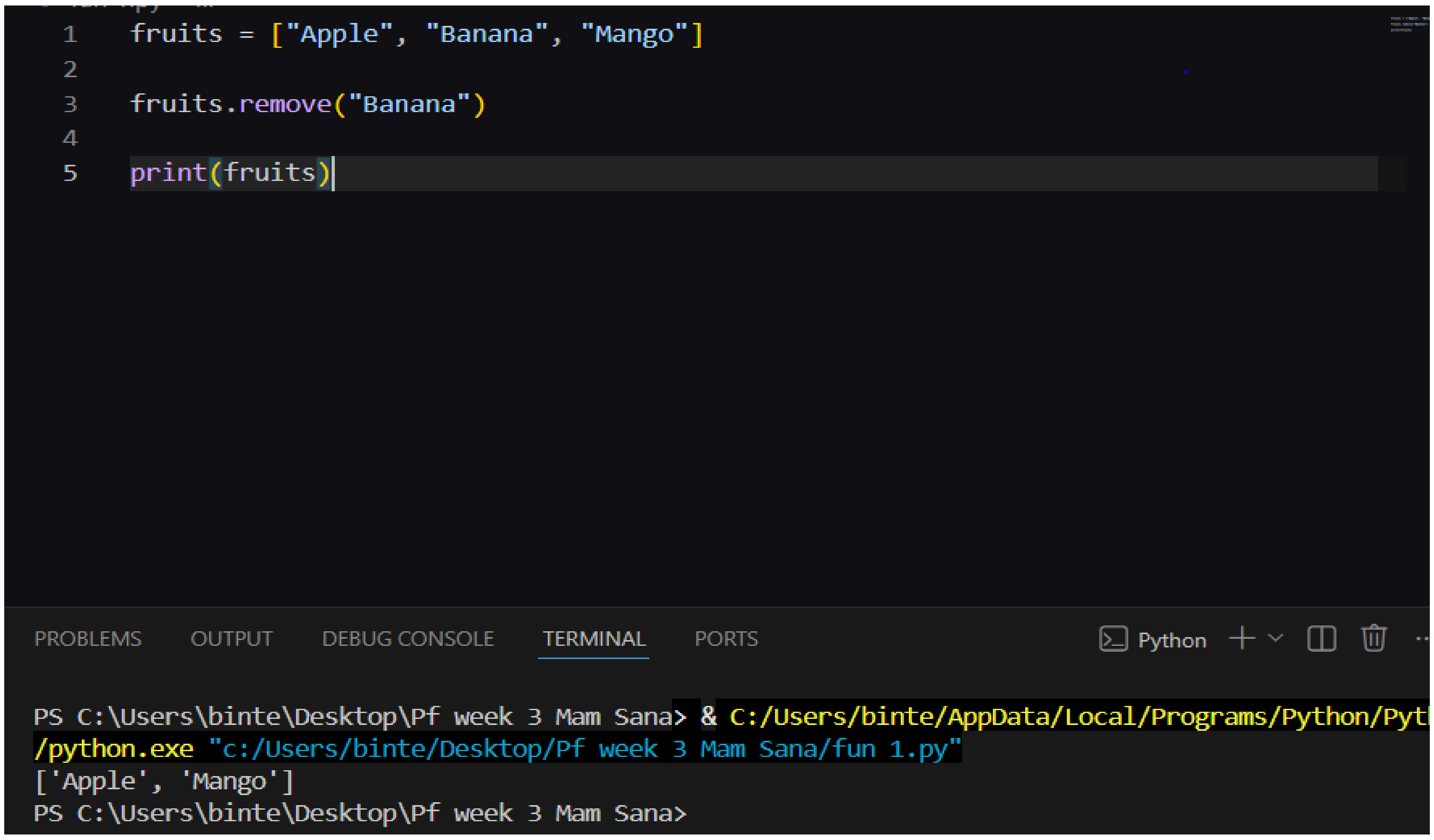
Task 2:


1 fruits = ["Apple", "Banana", "Mango"]
2
3 print(fruits[0])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Apple
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

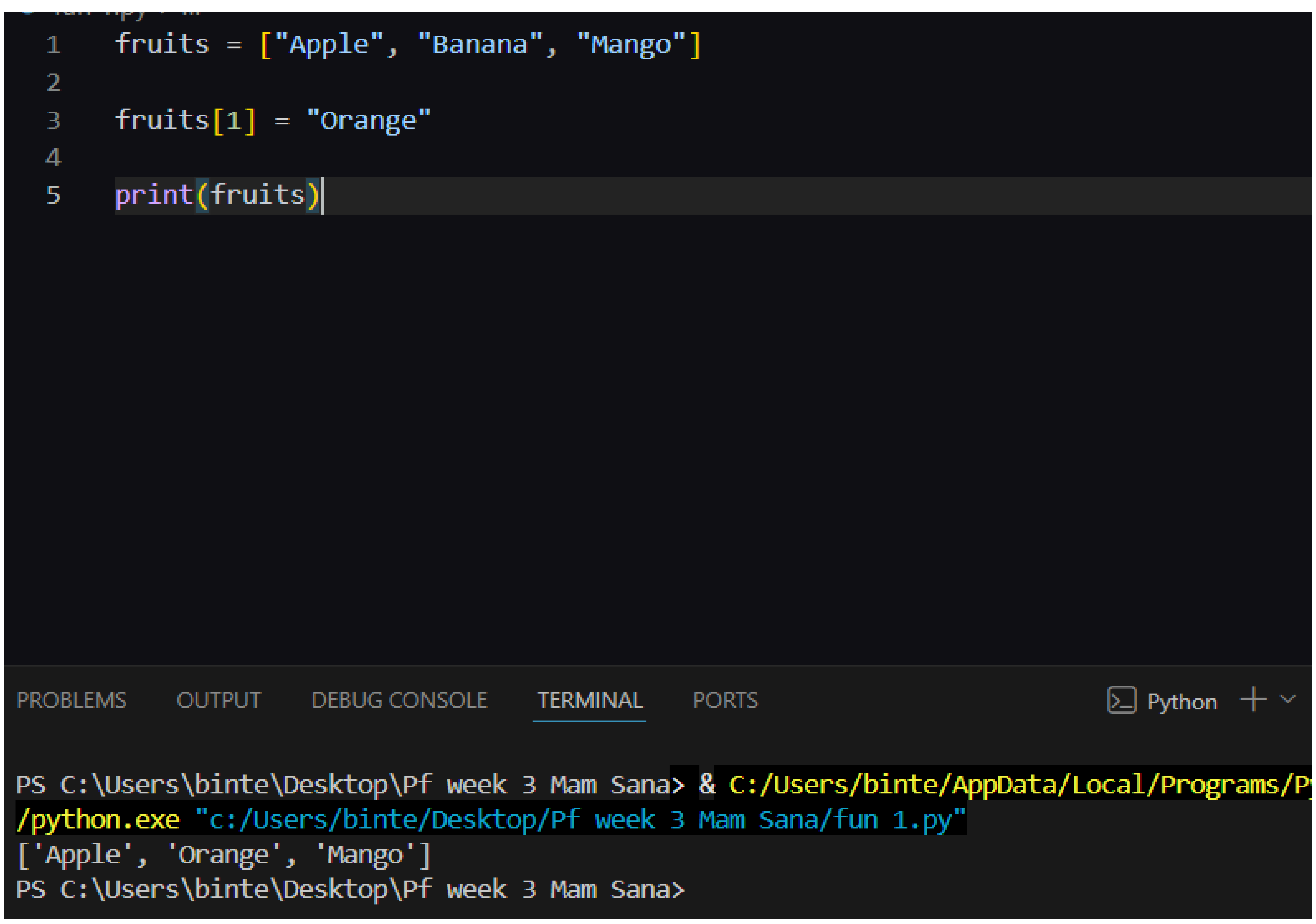
Task 3:


1 fruits = ["Apple", "Banana"]
2
3 fruits.append("Mango")
4
5 print(fruits)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Py
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
['Apple', 'Banana', 'Mango']
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

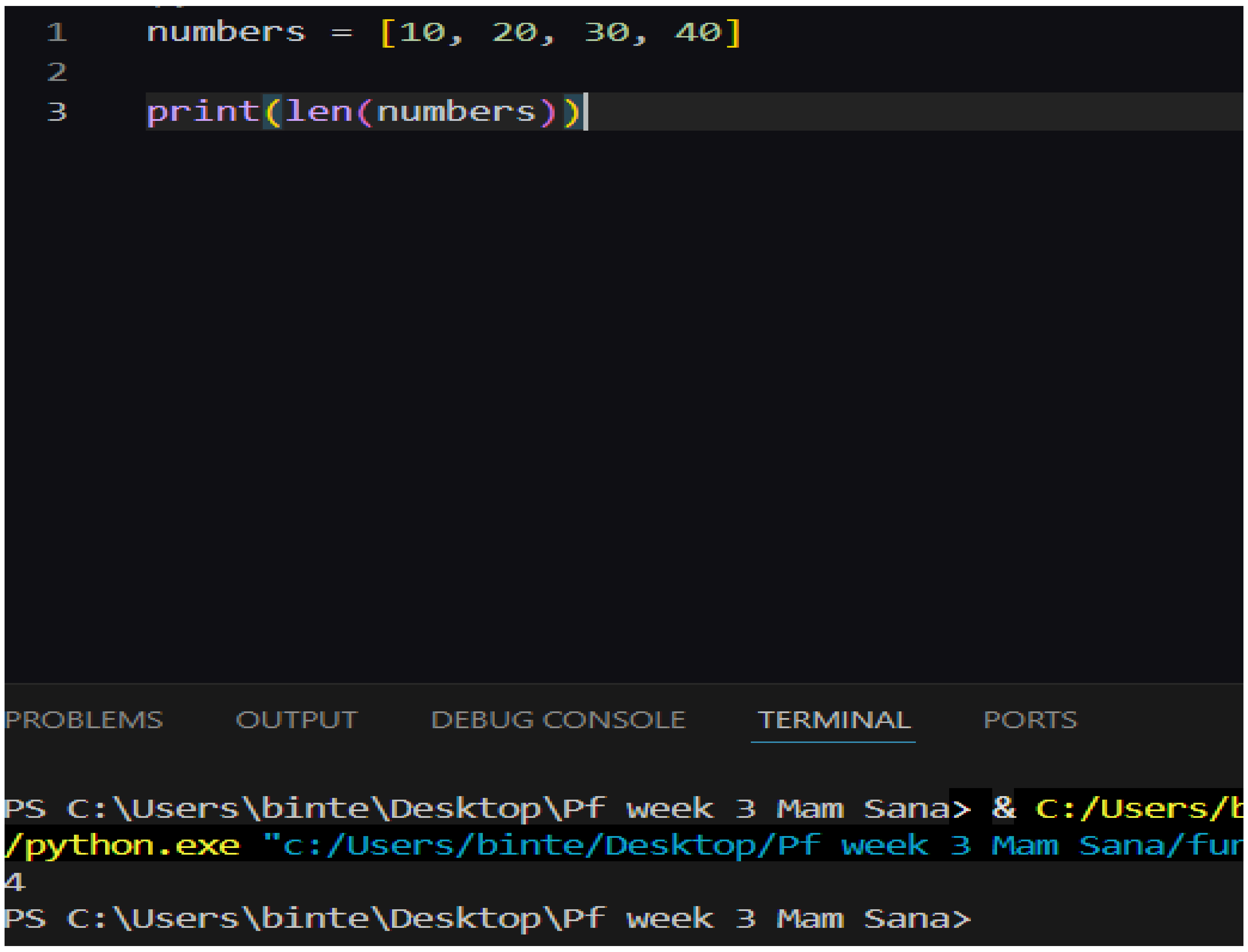
Task 4:


1 fruits = ["Apple", "Banana", "Mango"]
2
3 fruits.remove("Banana")
4
5 print(fruits)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pyt
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
['Apple', 'Mango']
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

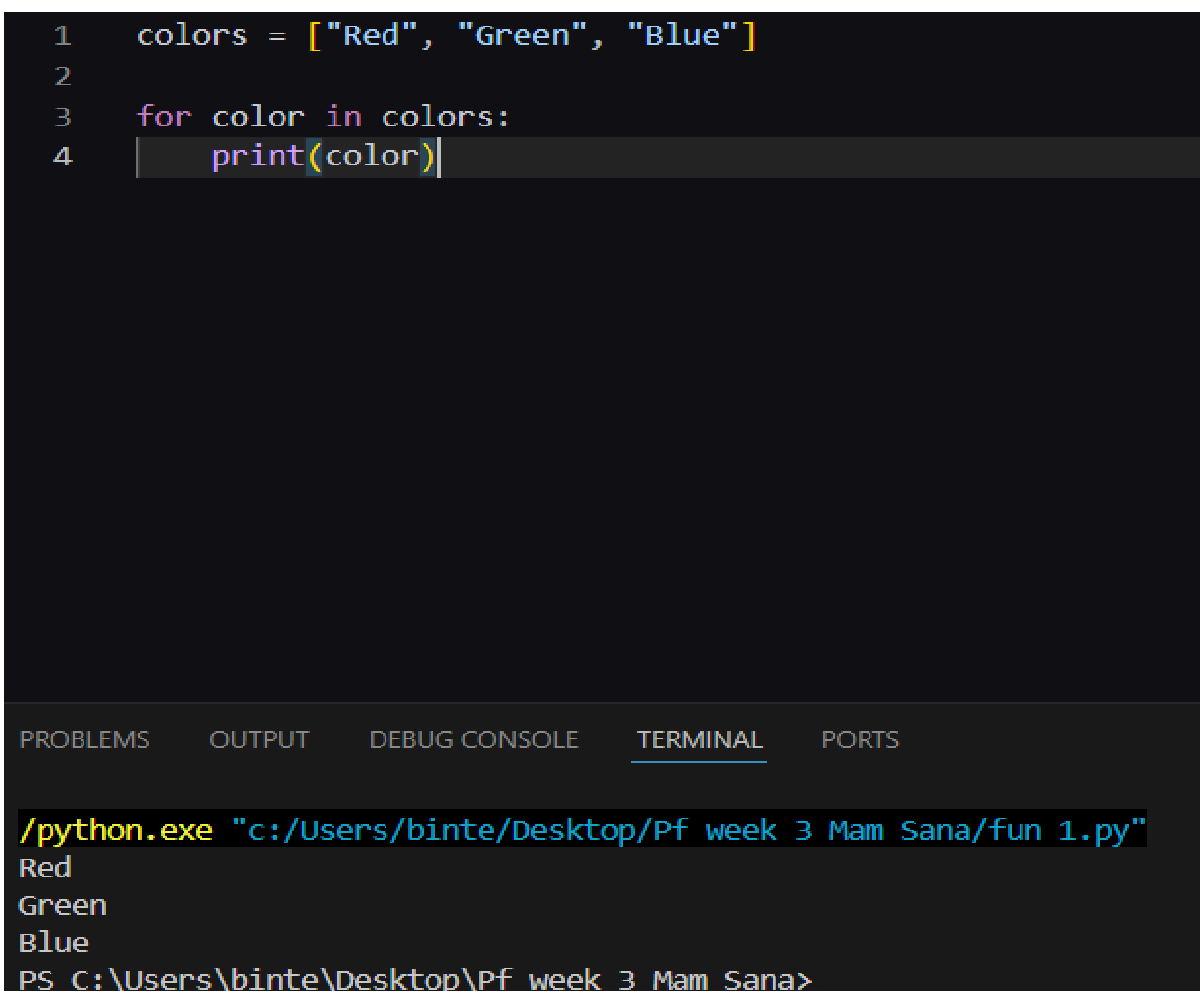
Task 5:


1 fruits = ["Apple", "Banana", "Mango"]
2
3 fruits[1] = "Orange"
4
5 print(fruits)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/P
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
['Apple', 'Orange', 'Mango']
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

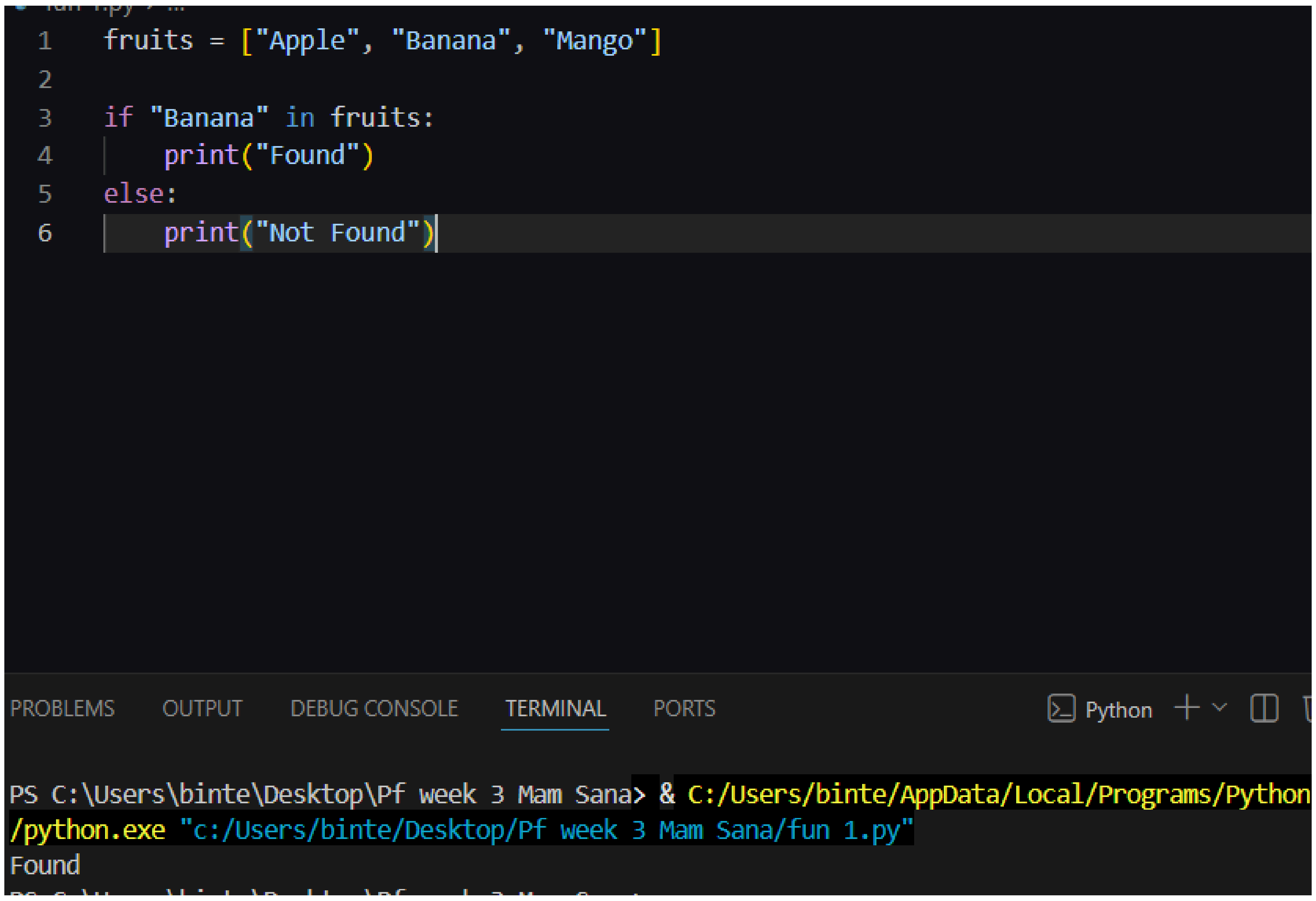
Task 6:


1 numbers = [10, 20, 30, 40]
2
3 print(len(numbers))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/b
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fur
4
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

Task 7:

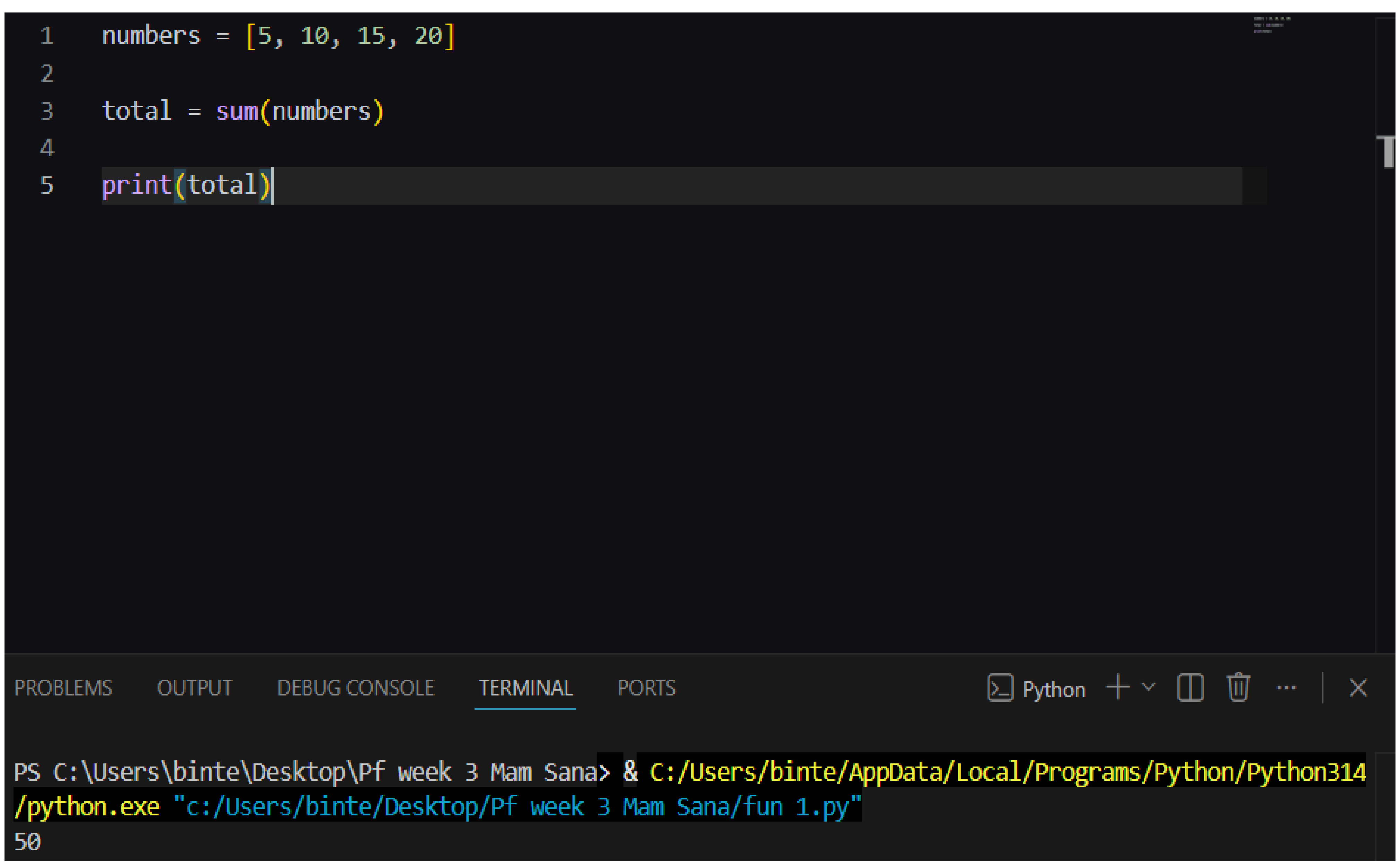

1 colors = ["Red", "Green", "Blue"]
2
3 for color in colors:
4   print(color)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Red
Green
Blue
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>

Task 8:



fruits = ["Apple", "Banana", "Mango"]
if "Banana" in fruits:
print("Found")
else:
print("Not Found")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
Found

Task 9:



numbers = [5, 10, 15, 20]
total = sum(numbers)
print(total)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python314
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
50

Task 10:

```python
numbers = []

for i in range(3):
    num = int(input("Enter number: "))
    numbers.append(num)

print(numbers)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
Enter number: 4
Enter number: 7
Enter number: 8
[4, 7, 8]
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 11:

```python
b=(1,"Maria","Jvaria")
print(b)
print(b[1])
print(b[0])
```

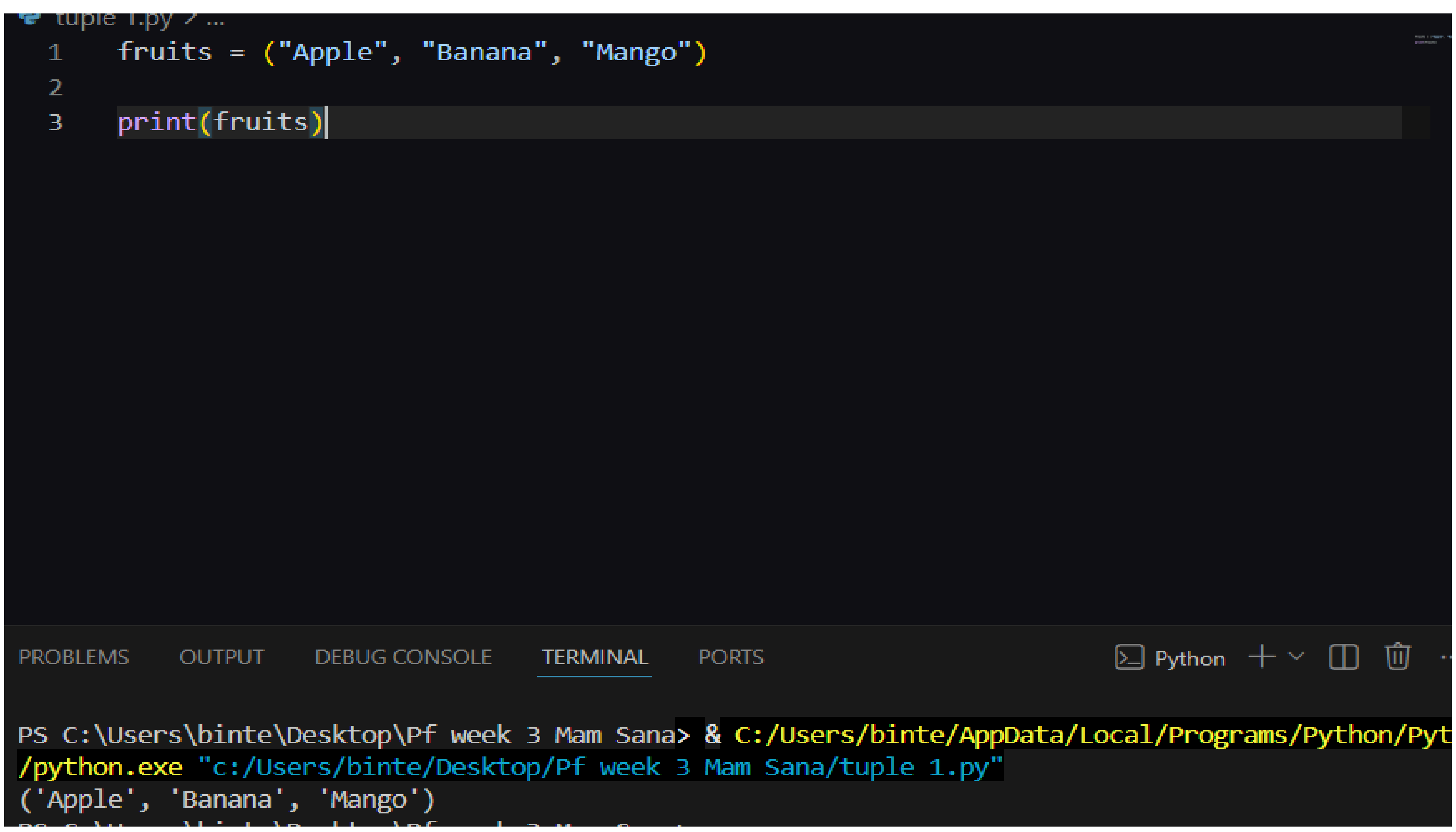
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Py

```
314/python.exe "c:/Users/binte/Desktop/List,tuple,dictionary/Task 2.py"
(1, 'Maria', 'Jvaria')
Maria
1
PS C:\Users\binte\Desktop\List,tuple,dictionary>
```

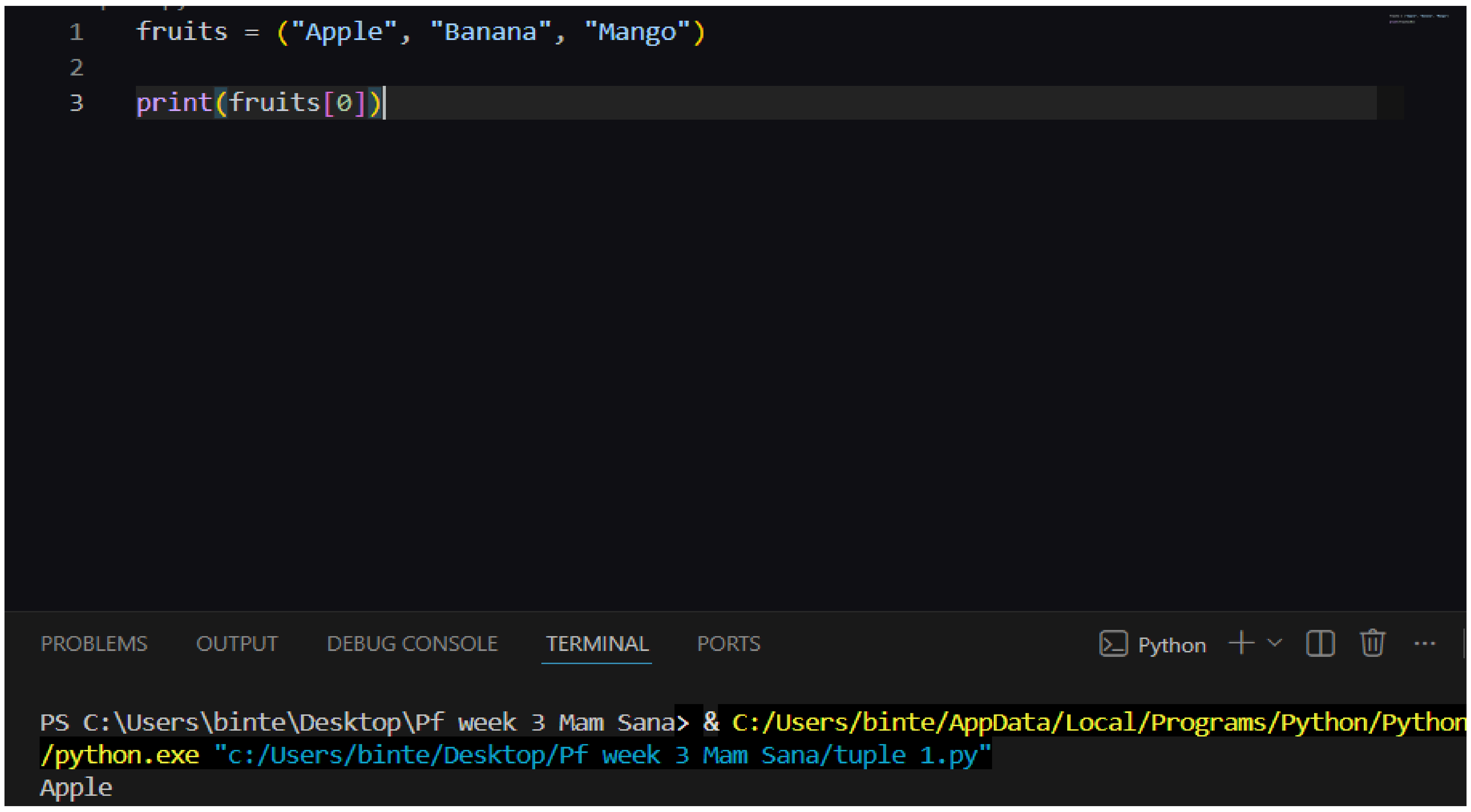
Task 12:

Tuples

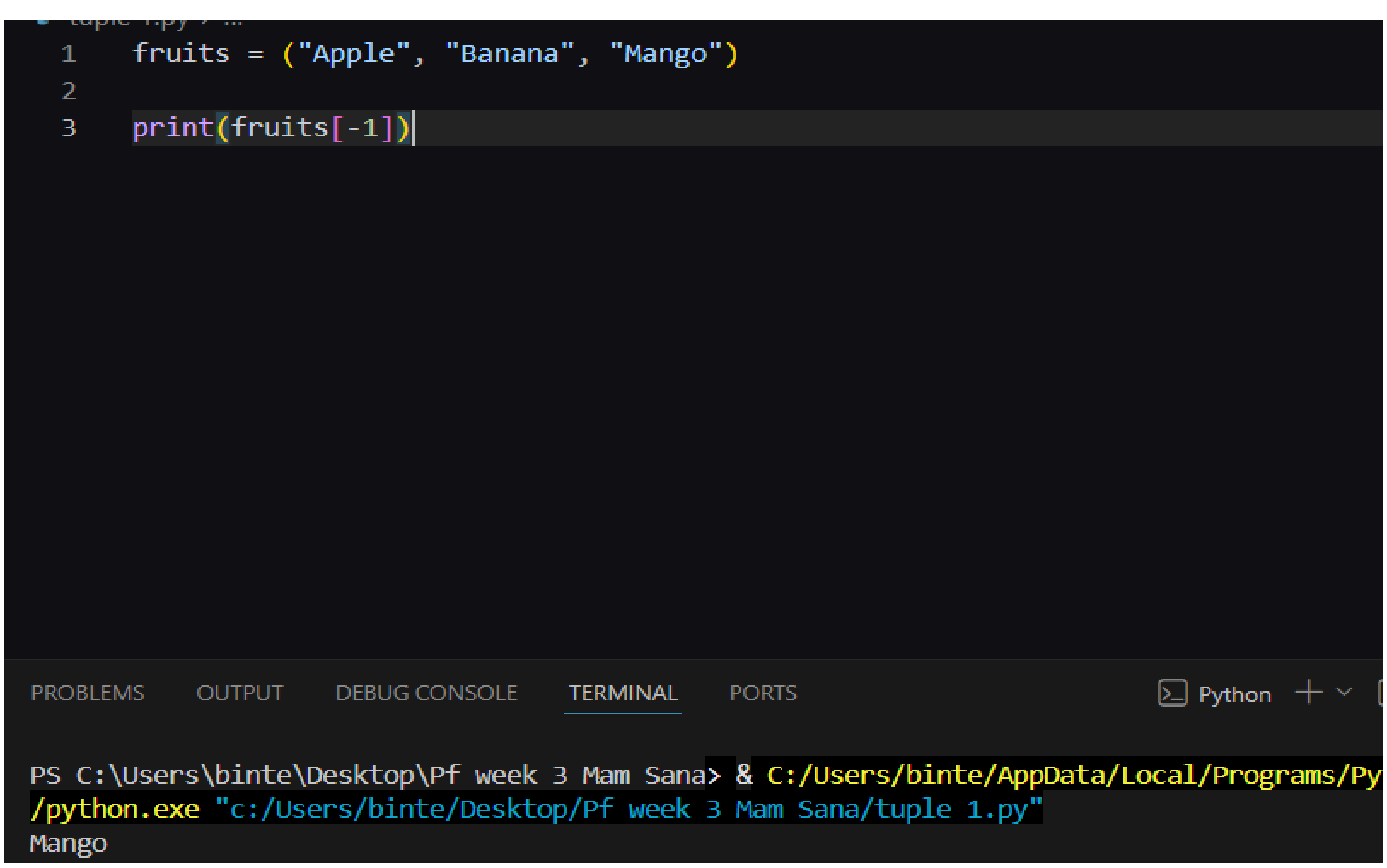
Task 1:



Task 2:


1 fruits = ("Apple", "Banana", "Mango")
2
3 print(fruits[0])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/tuple 1.py"
Apple

Task 3:


1 fruits = ("Apple", "Banana", "Mango")
2
3 print(fruits[-1])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Py
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/tuple 1.py"
Mango

Task 4:

```python
numbers = (10, 20, 30, 40)

print(len(numbers))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pyth
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/tuple 1.py"
4
```

Task 5:

```python
fruits = ("Apple", "Banana", "Mango")

my_list = list(fruits)

print(my_list)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/t
['Apple', 'Banana', 'Mango']
```

University of Engineering and Technology, Lahore

Name: Maria Abbas

Reg No: 2026(S) CYS_61

Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 8

Date: 4 May 2026

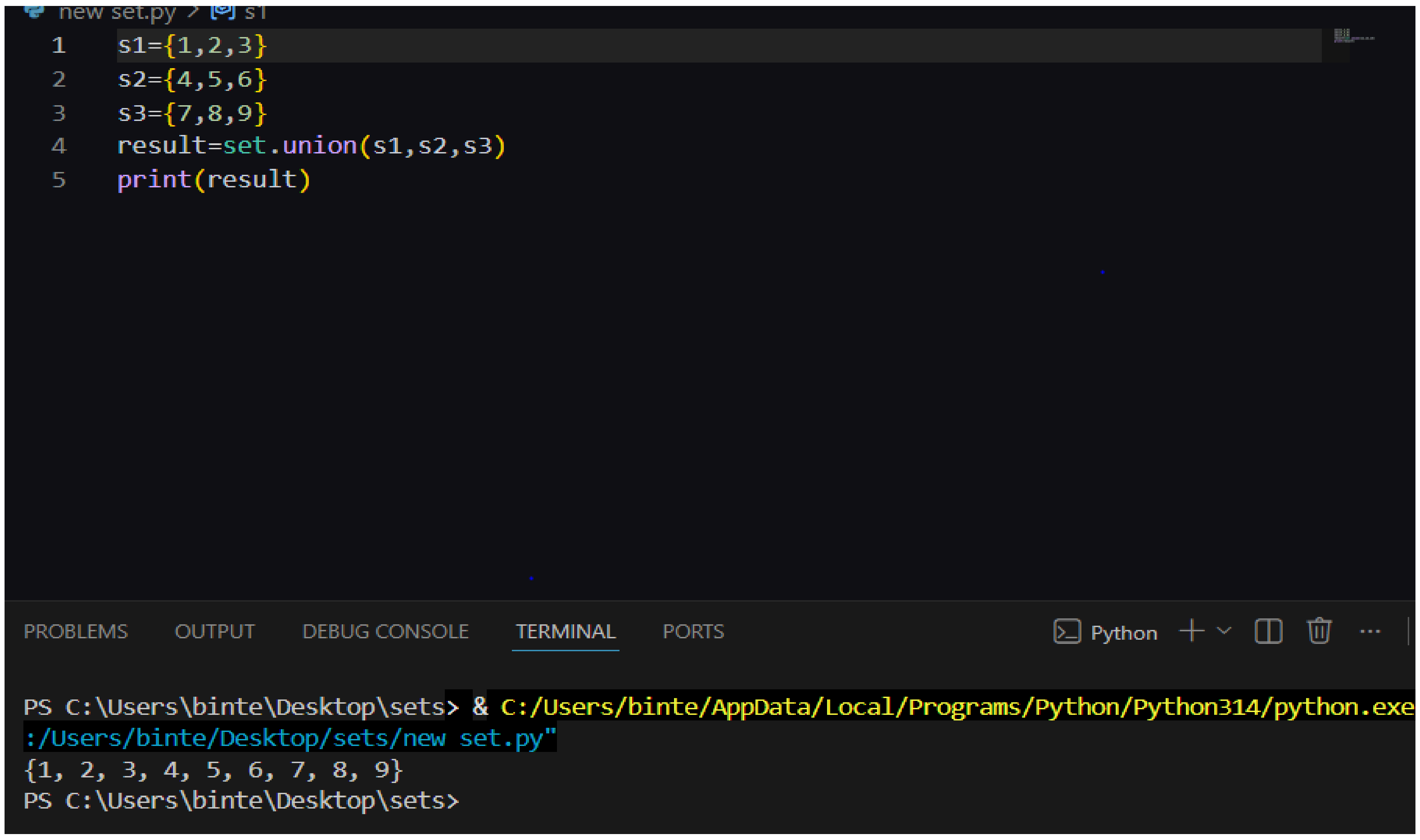
CLO3

Lab Tasks

Sets and Dictionary

Sets:

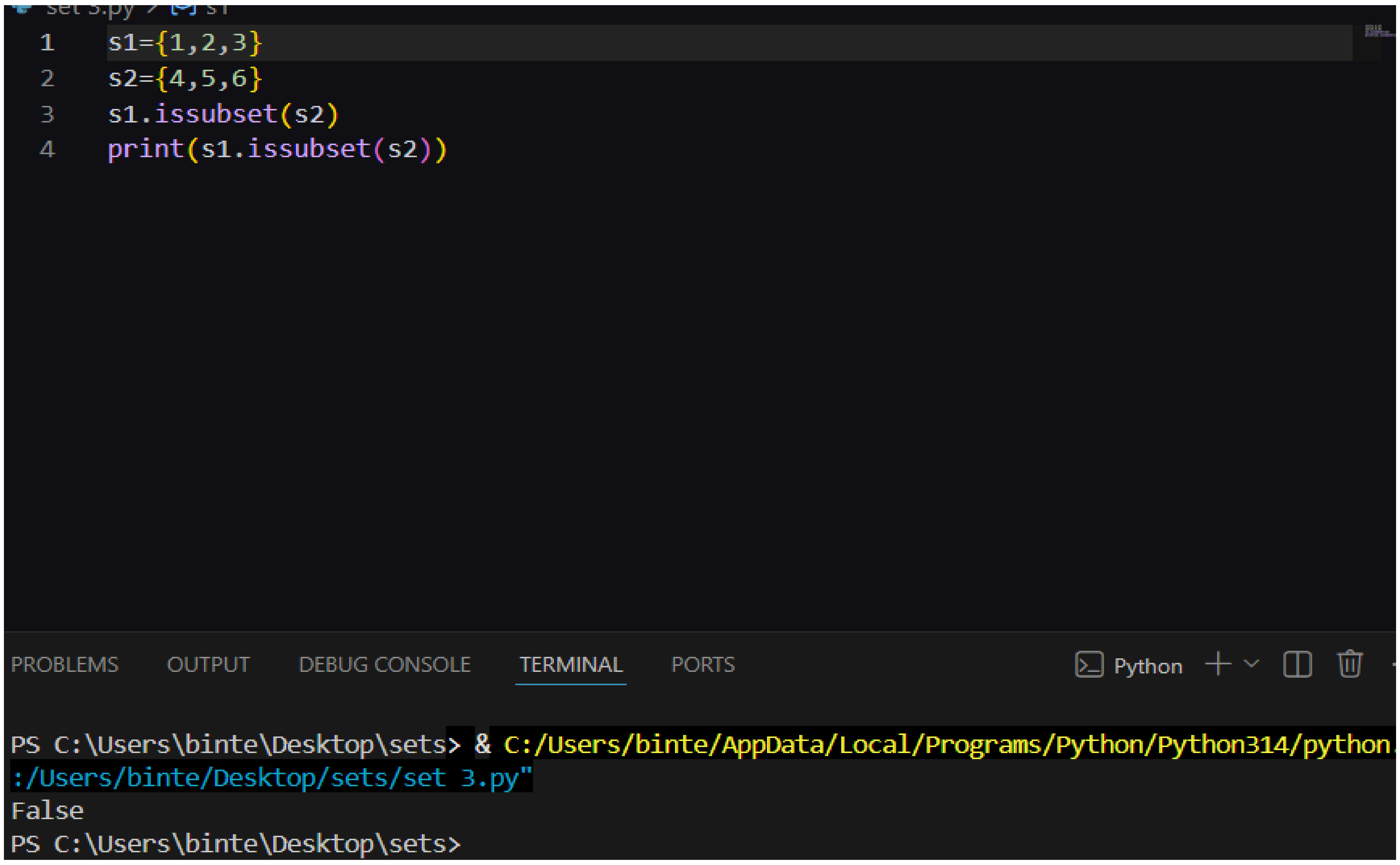
Task 1:


new set.py > s1
s1={1,2,3}
s2={4,5,6}
s3={7,8,9}
result=set.union(s1,s2,s3)
print(result)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
:/Users/binte/Desktop/sets/new set.py"
{1, 2, 3, 4, 5, 6, 7, 8, 9}
PS C:\Users\binte\Desktop\sets>

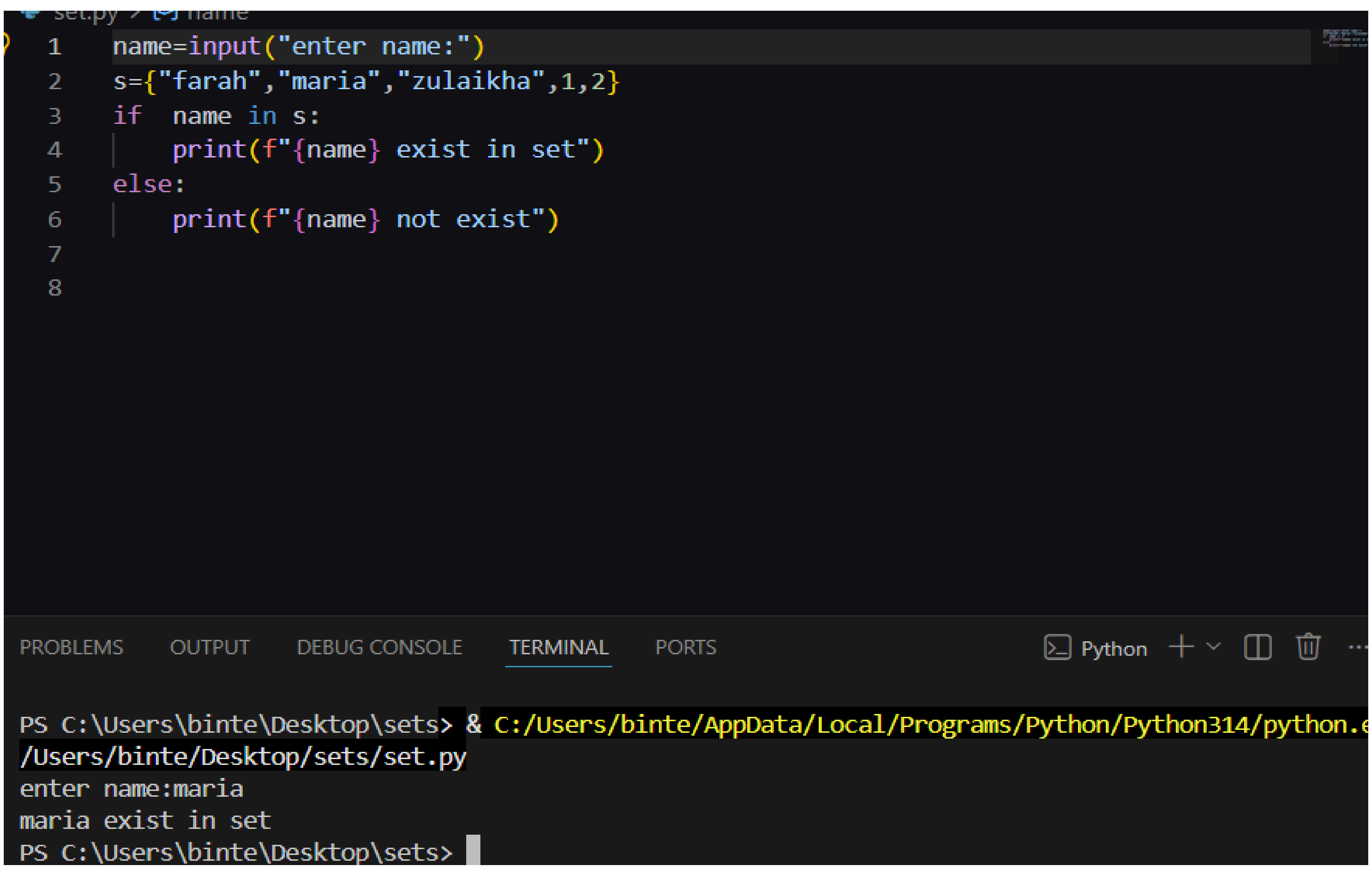
Task 2:


s={"farah","zulaikha", "maria"}
s.add("saima")
s.discard("zulaikha")
print(s)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.
:/Users/binte/Desktop/sets/set 1.py"
{'saima', 'maria', 'farah'}
PS C:\Users\binte\Desktop\sets>

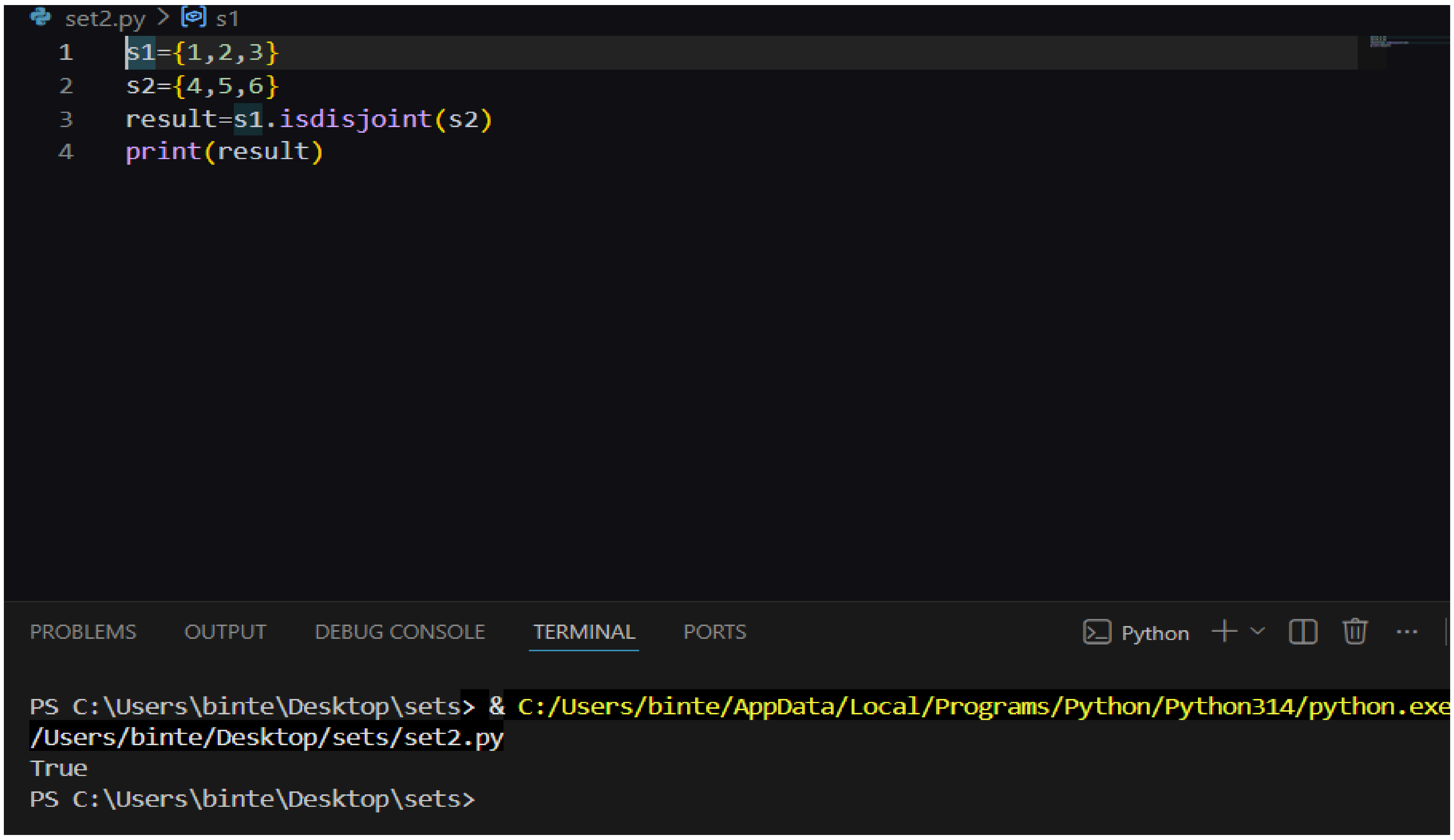
Task 3:


set 3.py > s1
s1={1,2,3}
s2={4,5,6}
s1.issubset(s2)
print(s1.issubset(s2))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.
:/Users/binte/Desktop/sets/set 3.py"
False
PS C:\Users\binte\Desktop\sets>

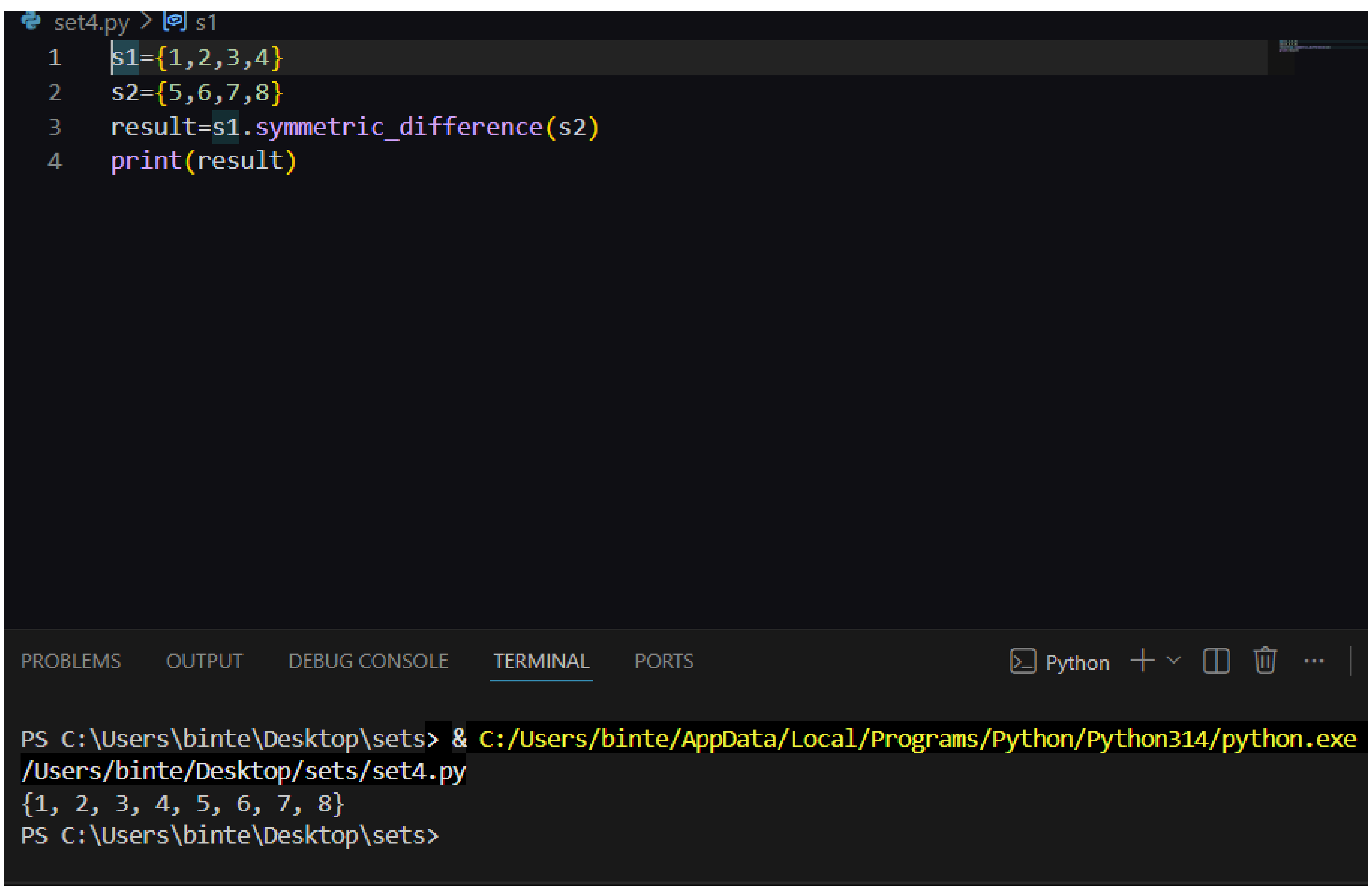
Task 4:


set.py > name
name=input("enter name:")
s={"farah","maria","zulaikha",1,2}
if name in s:
    print(f"{name} exist in set")
else:
    print(f"{name} not exist")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.e
/Users/binte/Desktop/sets/set.py
enter name:maria
maria exist in set
PS C:\Users\binte\Desktop\sets>

Task 5:


set2.py > s1
s1={1,2,3}
s2={4,5,6}
result=s1.isdisjoint(s2)
print(result)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe /Users/binte/Desktop/sets/set2.py
True
PS C:\Users\binte\Desktop\sets>

Task 6:


set4.py > s1
s1={1,2,3,4}
s2={5,6,7,8}
result=s1.symmetric_difference(s2)
print(result)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe /Users/binte/Desktop/sets/set4.py
{1, 2, 3, 4, 5, 6, 7, 8}
PS C:\Users\binte\Desktop\sets>

Task 7:

```python
s1={1,2,3,4}
s2={5,6,7,8}
s3={9,10,}
result=s1.update(s2,s3)
print(s1)
```

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe c:
/Users/binte/Desktop/sets/set5.py
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
PS C:\Users\binte\Desktop\sets>
```

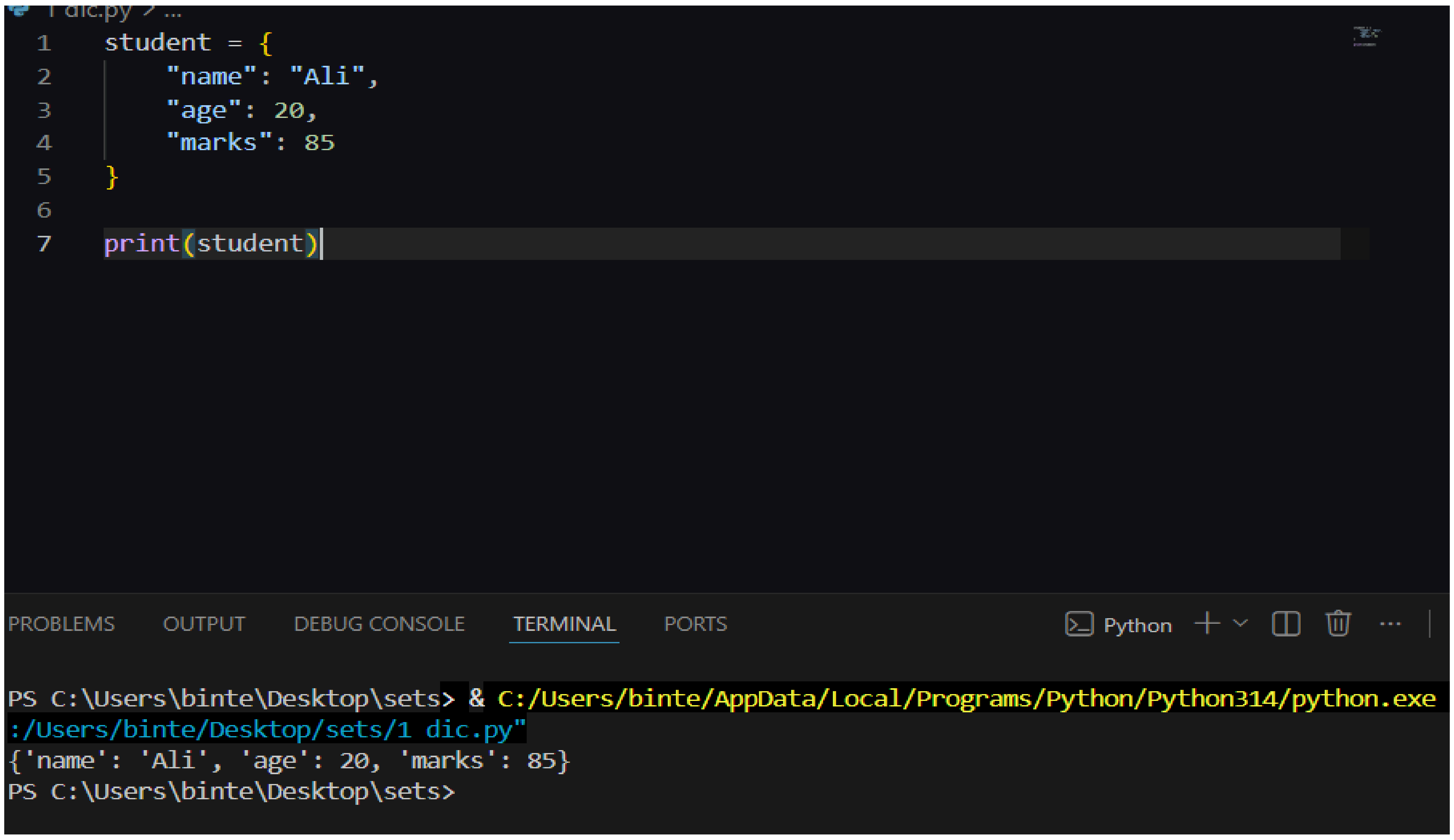
Task 8:

```python
s1={1,2,3,4,8}
s2={5,6,7,8}
s3={8,9,10,}
result=s1.intersection_update(s2,s3)
print(s1)
```

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
/Users/binte/Desktop/sets/set6.py
{8}
PS C:\Users\binte\Desktop\sets>
```

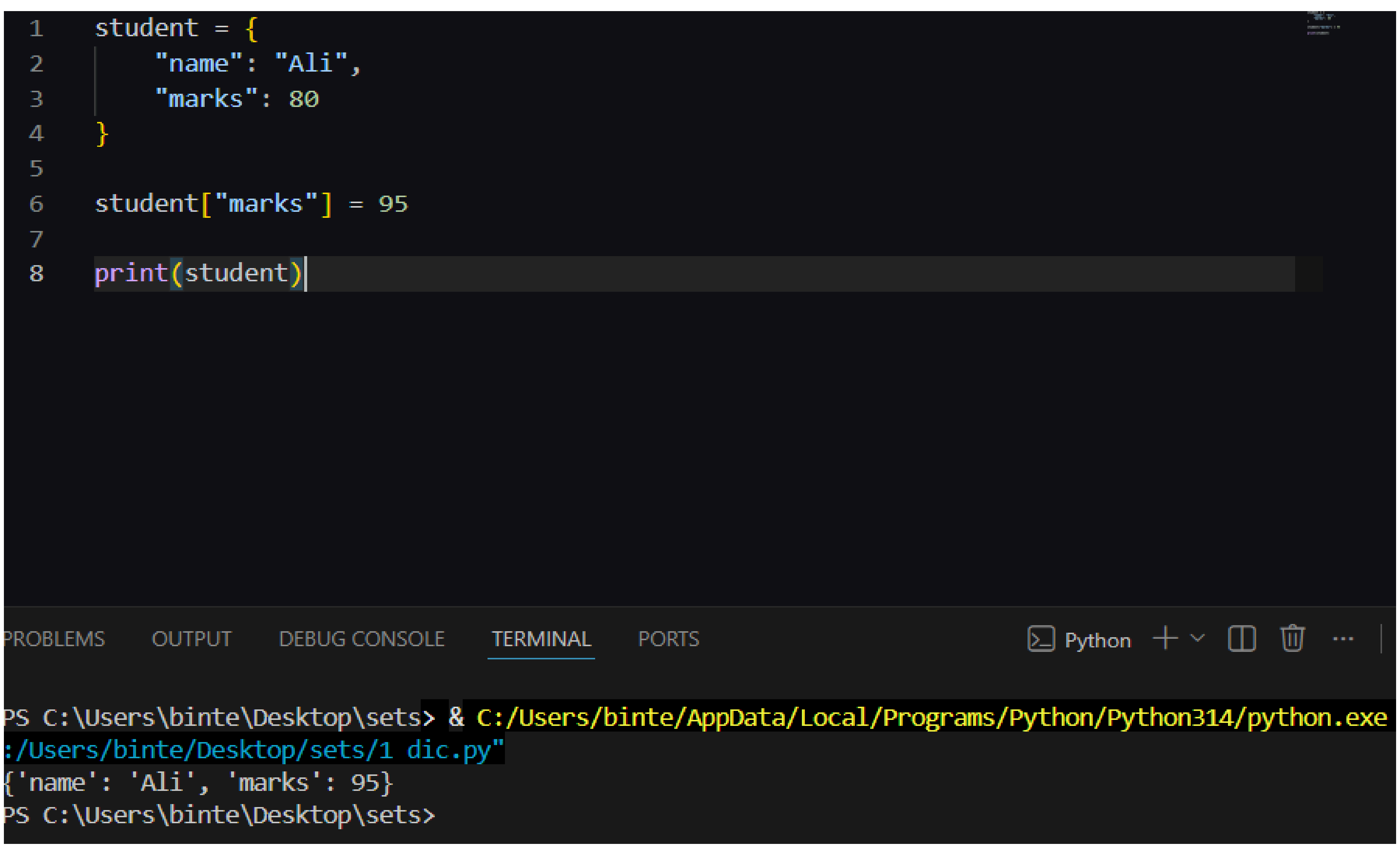
Dictionary

Task 1:



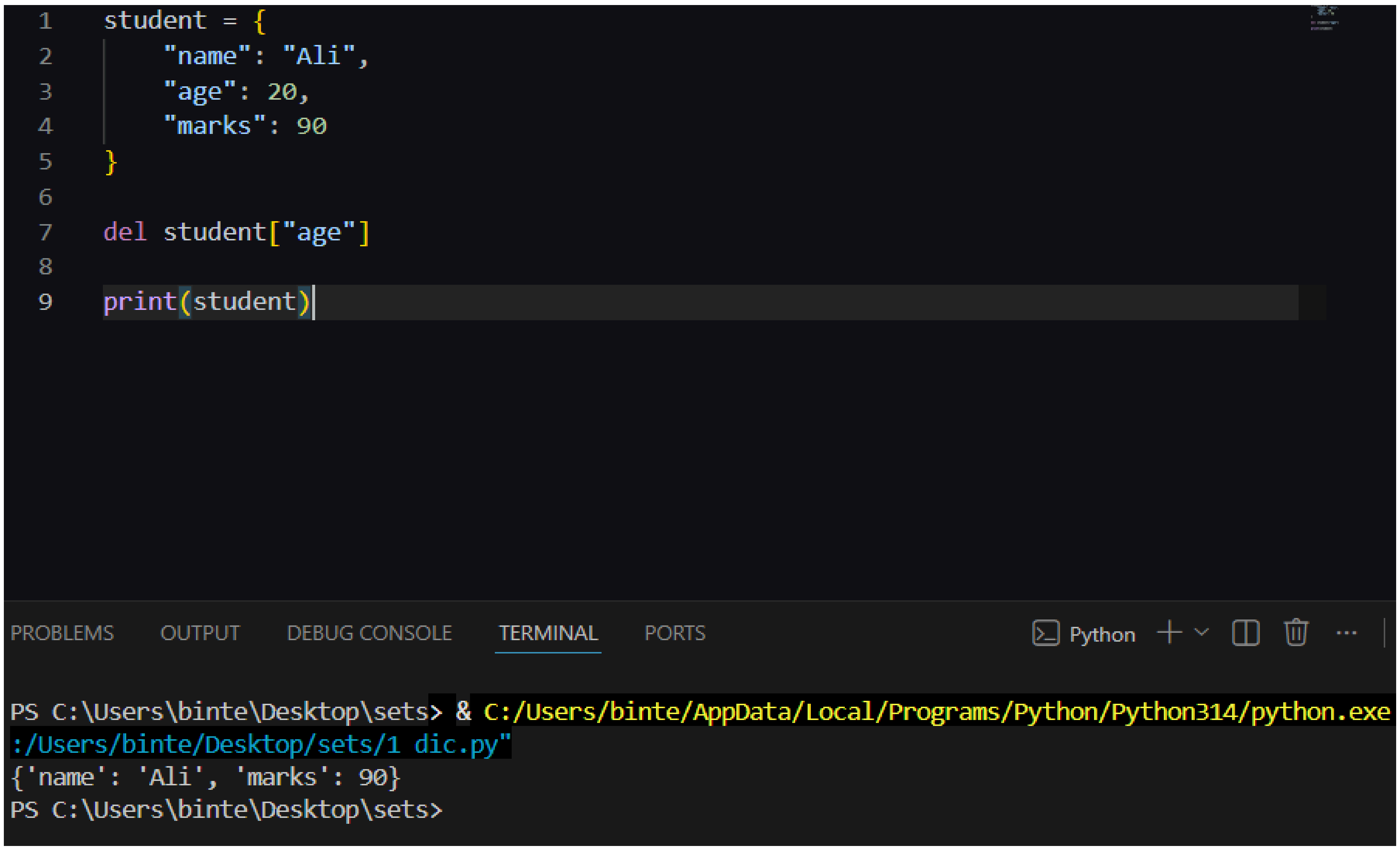
1 dic.py > ...
student = {
"name": "Ali",
"age": 20,
"marks": 85
}
print(student)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
:/Users/binte/Desktop/sets/1 dic.py"
{'name': 'Ali', 'age': 20, 'marks': 85}
PS C:\Users\binte\Desktop\sets>

Task 2:



student = {
"name": "Ali",
"marks": 80
}
student["marks"] = 95
print(student)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
:/Users/binte/Desktop/sets/1 dic.py"
{'name': 'Ali', 'marks': 95}
PS C:\Users\binte\Desktop\sets>

Task 3:



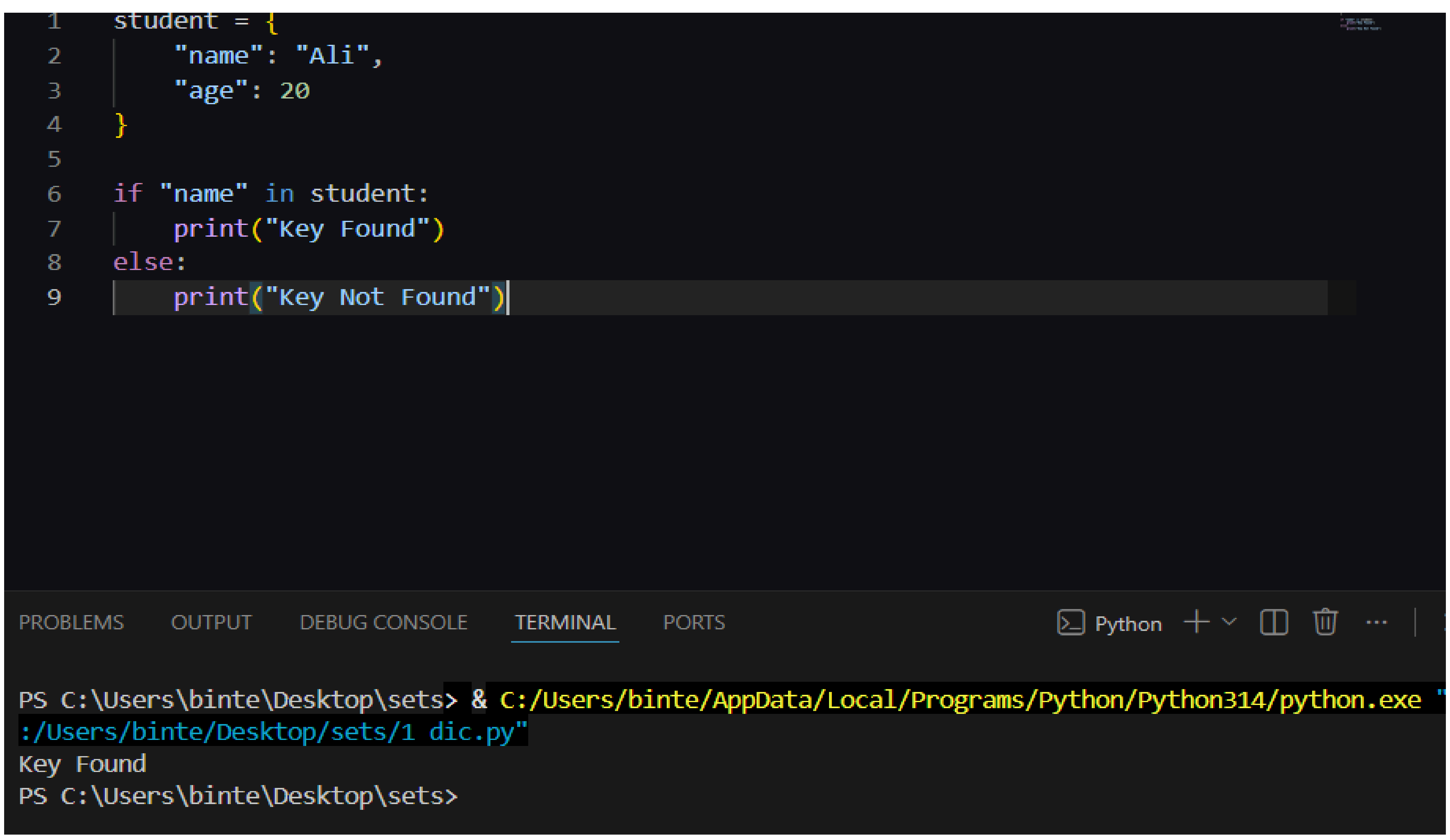
```python
student = {
    "name": "Ali",
    "age": 20,
    "marks": 90
}

del student["age"]

print(student)
```

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
:/Users/binte/Desktop/sets/1 dic.py"
{'name': 'Ali', 'marks': 90}
PS C:\Users\binte\Desktop\sets>
```

Task 4:



```python
student = {
    "name": "Ali",
    "age": 20
}

if "name" in student:
    print("Key Found")
else:
    print("Key Not Found")
```

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe "
:/Users/binte/Desktop/sets/1 dic.py"
Key Found
PS C:\Users\binte\Desktop\sets>
```

Task 5:

```python
student = {}

student["name"] = input("Enter Name: ")
student["age"] = int(input("Enter Age: "))

print(student)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
:/Users/binte/Desktop/sets/1 dic.py"
Enter Name: maria
Enter Age: 19
{'name': 'maria', 'age': 19}
PS C:\Users\binte\Desktop\sets>
```

University of Engineering and Technology, Lahore

Name: Maria Abbas

Reg No: 2026(S) CYS_61

Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 10

Date: 18 May 2026

CLO2

Lab Tasks

Recursion

Task 1:

```python
def print_numbers(n):
    if n == 0:
        return
    print_numbers(n - 1)
    print(n)

print_numbers(5)
```

```
2
3
4
5
```

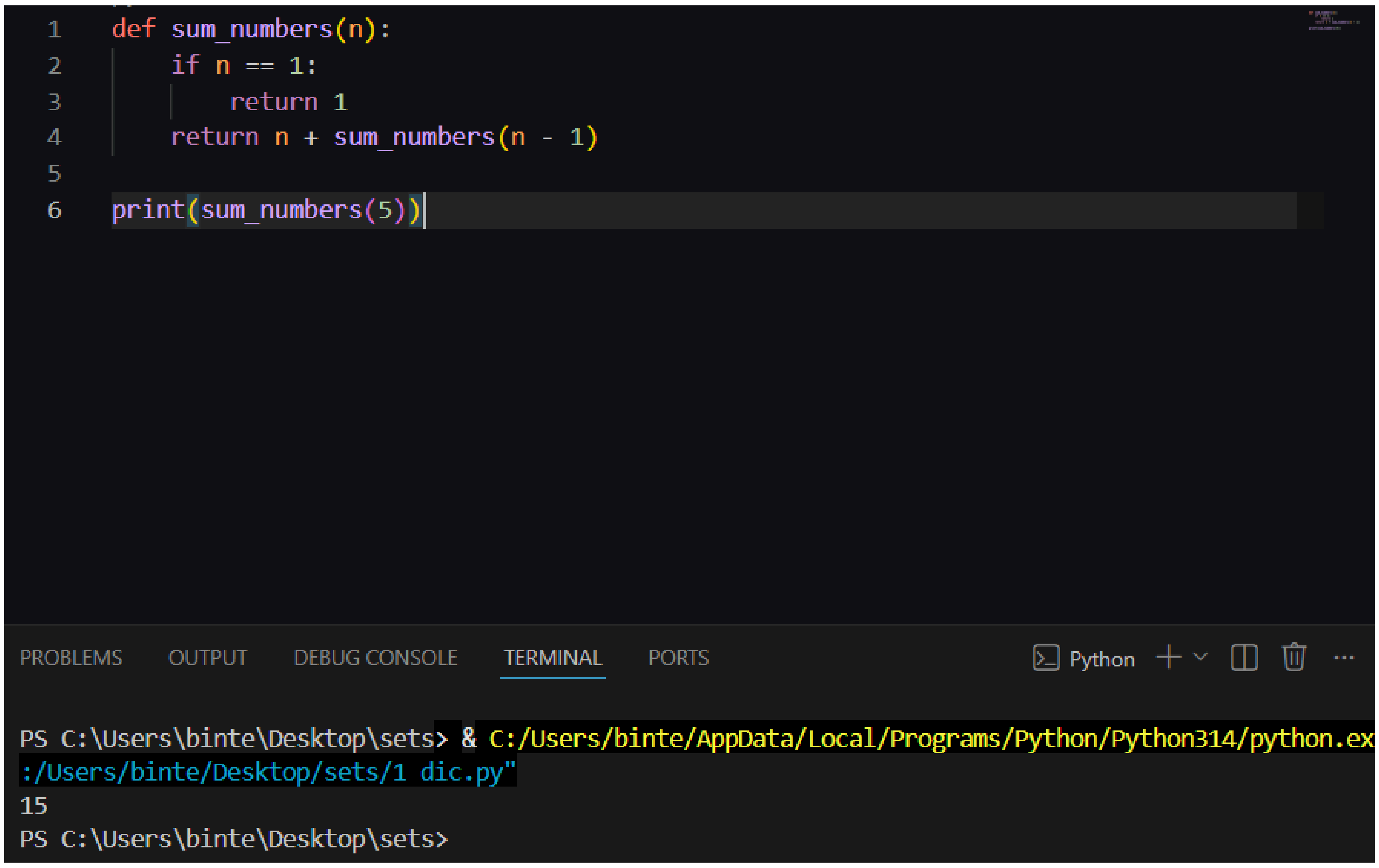
Task 2:

```python
def countdown(n):
    if n == 0:
        return
    print(n)
    countdown(n - 1)

countdown(5)
```

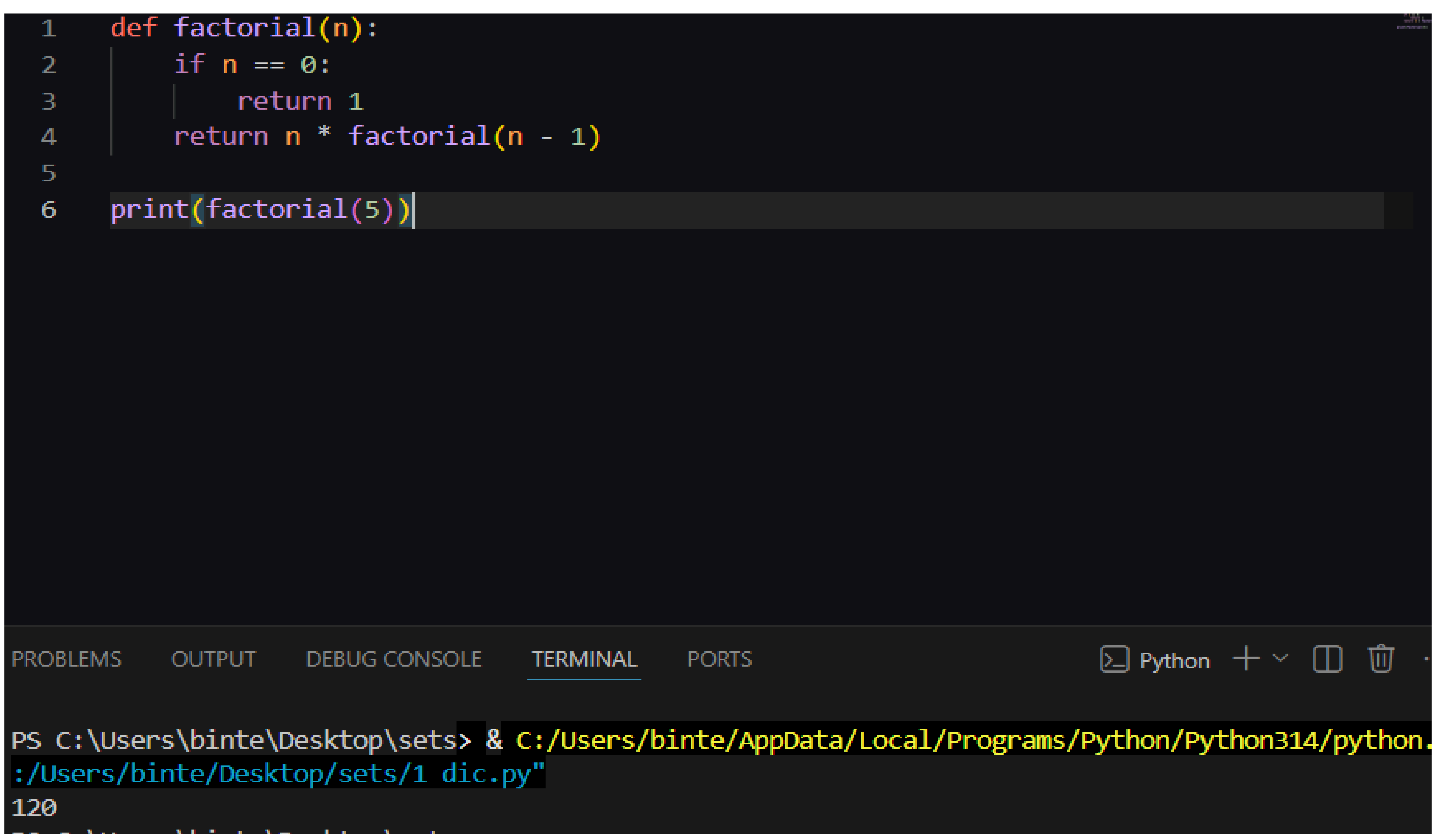
```
4
3
2
1
```

Task 3:


def sum_numbers(n):
    if n == 1:
        return 1
    return n + sum_numbers(n - 1)

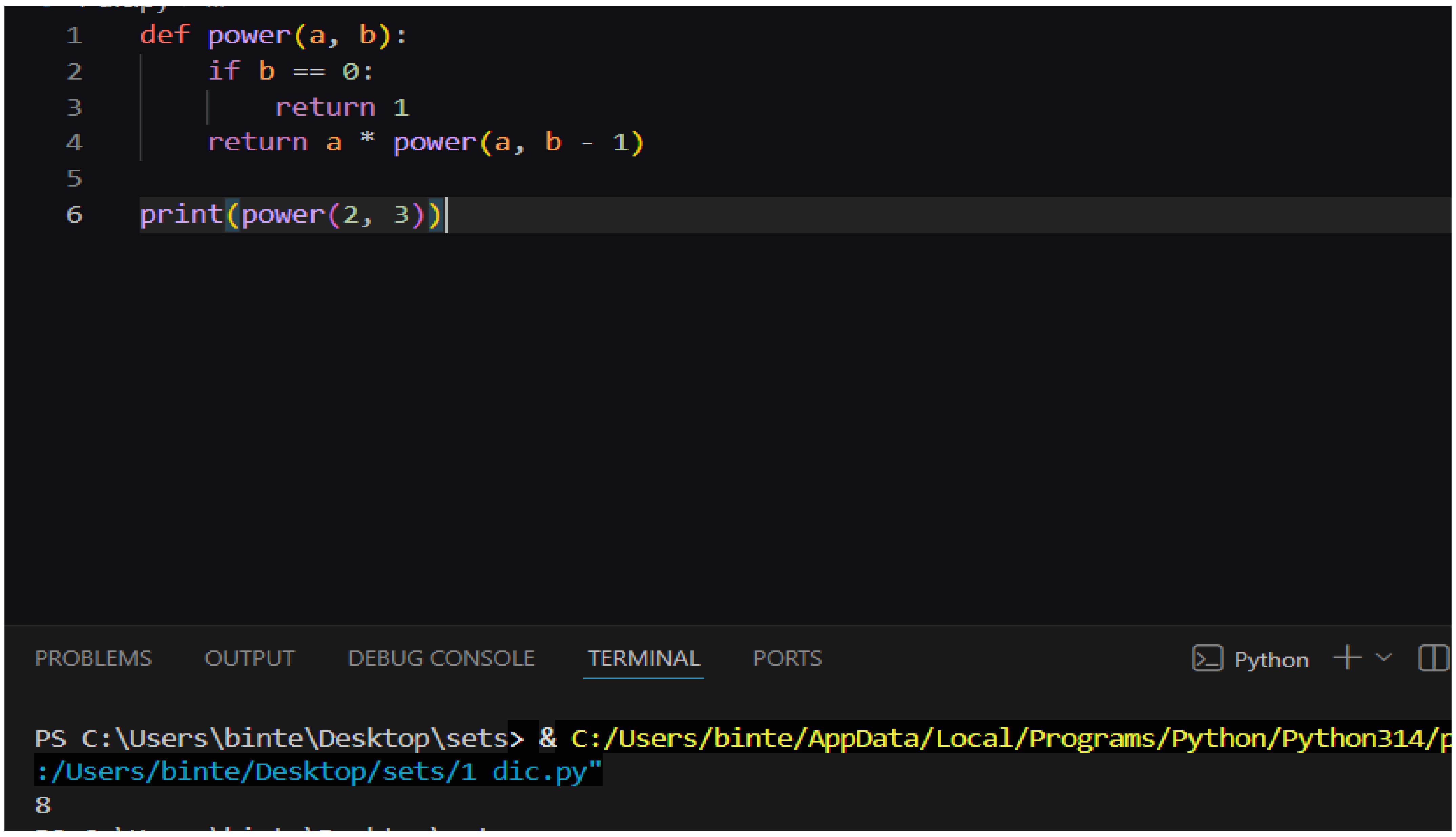
print(sum_numbers(5))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.ex
:/Users/binte/Desktop/sets/1 dic.py"
15
PS C:\Users\binte\Desktop\sets>

Task 4:

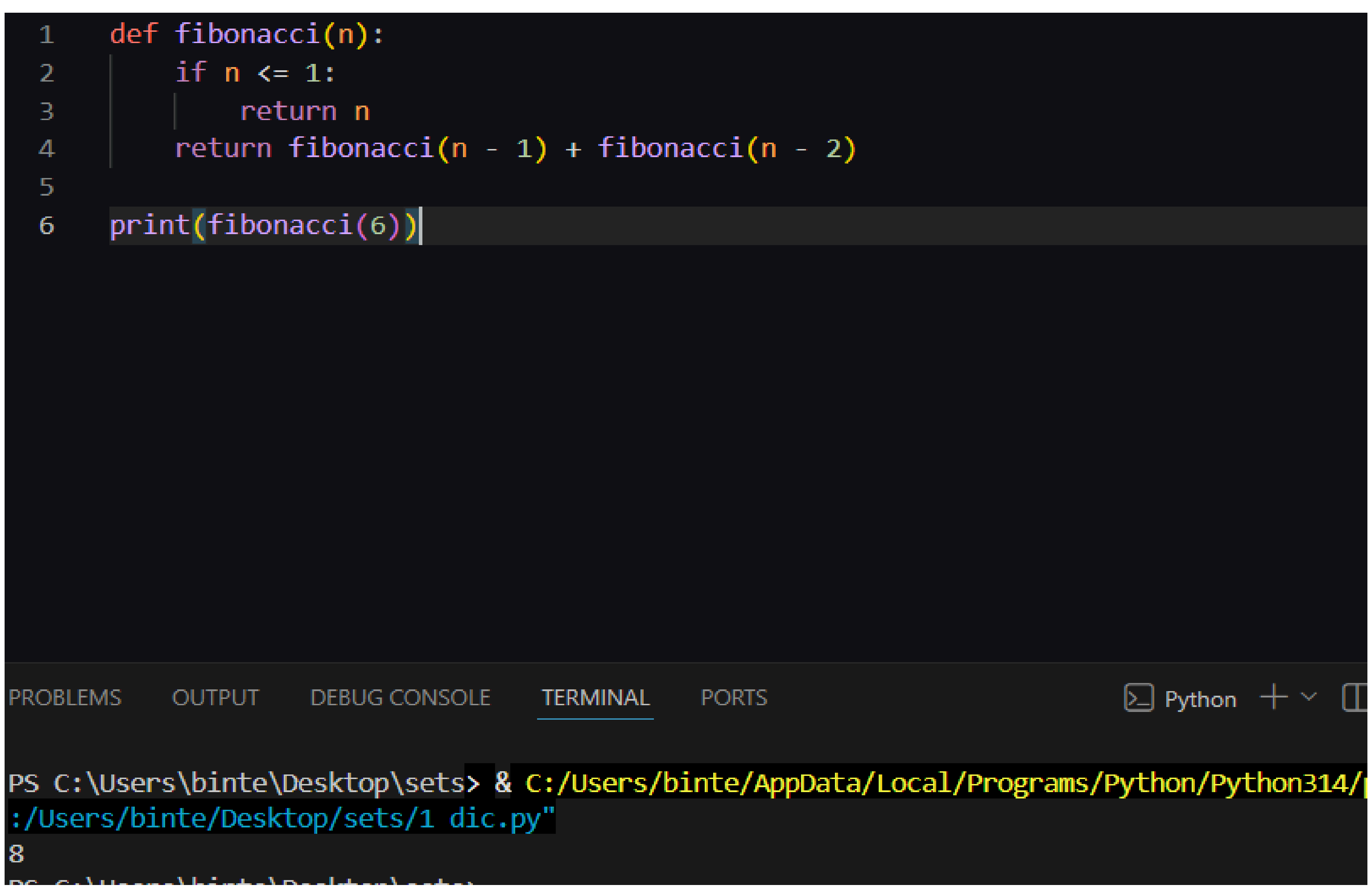

def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)

print(factorial(5))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.
:/Users/binte/Desktop/sets/1 dic.py"
120

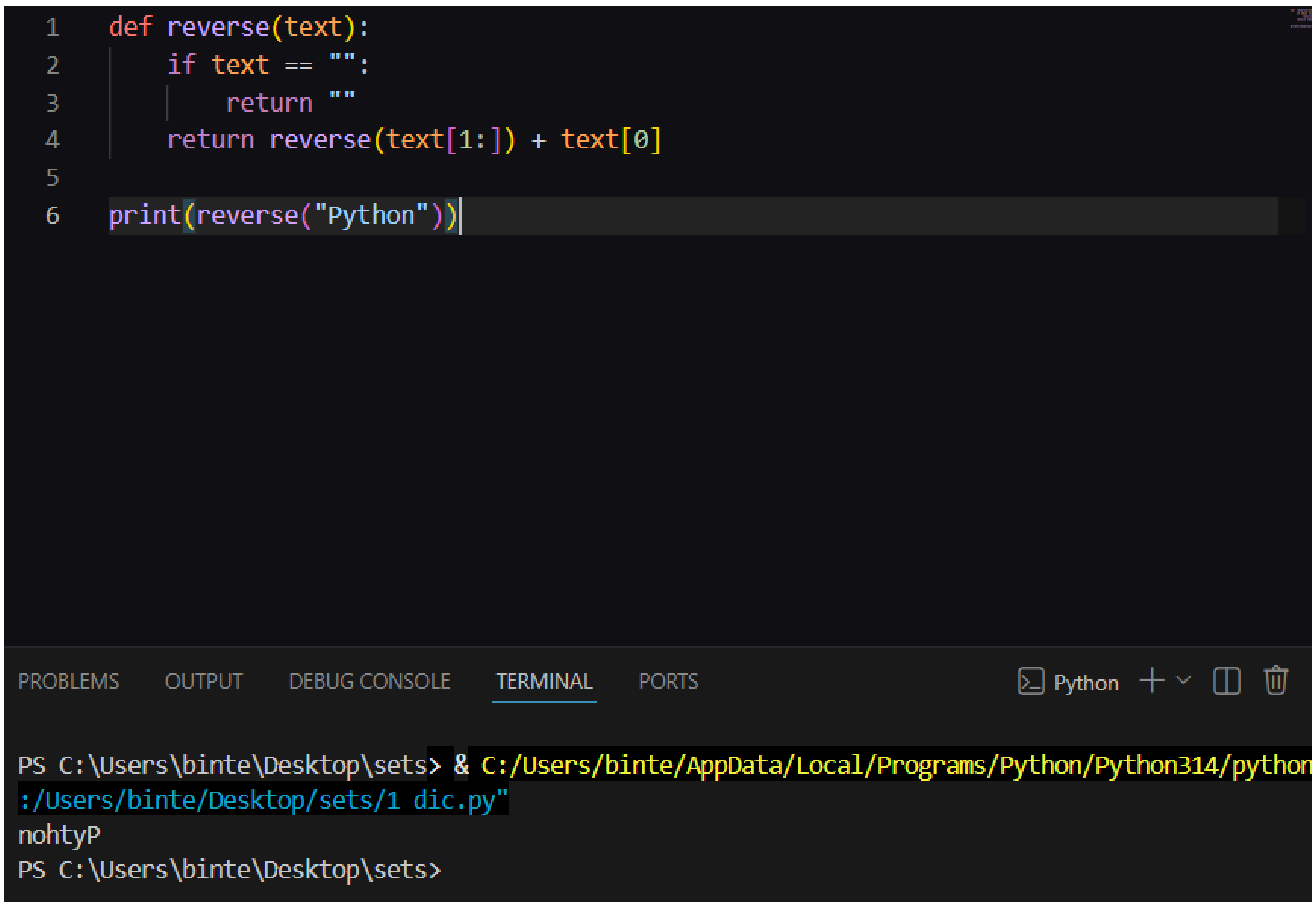
Task 5:


def power(a, b):
if b == 0:
return 1
return a * power(a, b - 1)
print(power(2, 3))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/
:/Users/binte/Desktop/sets/1 dic.py"
8

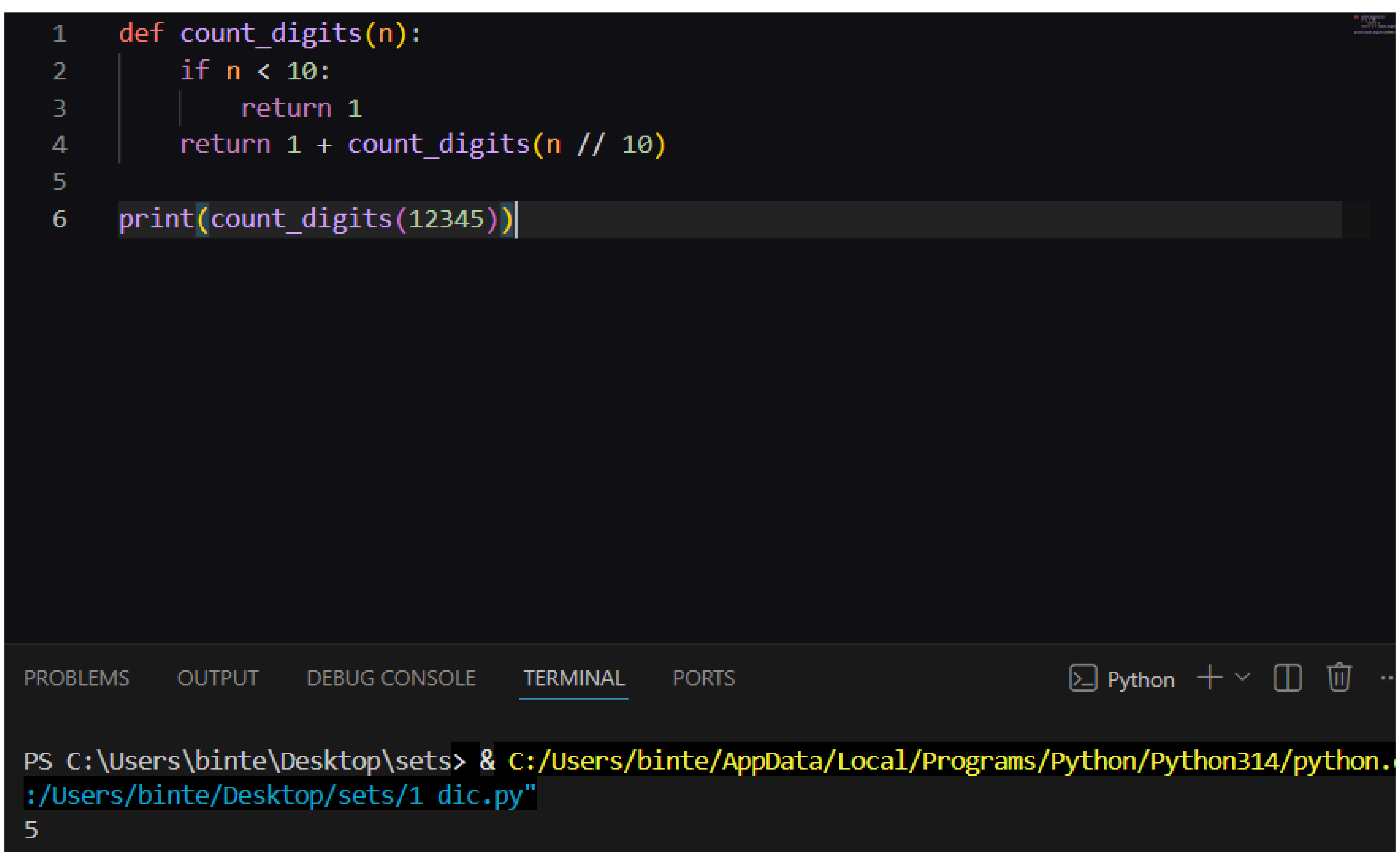
Task 6:


def fibonacci(n):
if n <= 1:
return n
return fibonacci(n - 1) + fibonacci(n - 2)
print(fibonacci(6))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/
:/Users/binte/Desktop/sets/1 dic.py"
8

Task 7:


def reverse(text):
if text == "":
return ""
return reverse(text[1:]) + text[0]
print(reverse("Python"))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python
:/Users/binte/Desktop/sets/1 dic.py"
nohtyP
PS C:\Users\binte\Desktop\sets>

Task8:


def count_digits(n):
if n < 10:
return 1
return 1 + count_digits(n // 10)
print(count_digits(12345))
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.
:/Users/binte/Desktop/sets/1 dic.py"
5

Task 9:

```python
def hello(n):
    if n == 0:
        return
    print("Hello")
    hello(n - 1)

hello(3)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
:/Users/binte/Desktop/sets/1 dic.py"
Hello
Hello
Hello
```

University of Engineering and Technology, Lahore

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Department: Computer Engineering

Degree: Cyber security

Section: B

Assignment: PF Lab week 11

Date: 18 May 2026

CLO4

Lab Tasks

Exception Handling

Task 1:

```python
try:
    num = 10 / 0
    print(num)
except:
    print("An error occurred.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314
:/Users/binte/Desktop/sets/1 dic.py"
An error occurred.
```

Task 2:

```python
try:
    age = int(input("Enter Age: "))
    print(age)
except ValueError:
    print("Please enter numbers only.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/
:/Users/binte/Desktop/sets/1 dic.py"
Enter Age: 19
19
```

Task 3:

```python
try:
    numbers = [10, 20, 30]
    print(numbers[5])
except IndexError:
    print("Index does not exist.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\sets> & C:/Users/binte/AppData/Local/Programs/Python/Python314/python.exe
:/Users/binte/Desktop/sets/1 dic.py"
Index does not exist.
```

Task 4:

```python
try:
    a = int(input("Enter First Number: "))
    b = int(input("Enter Second Number: "))
    print(a / b)
except Exception as e:
    print("Error:", e)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
:/Users/binte/Desktop/sets/1 dic.py"
Enter First Number: 3
Enter Second Number: 4
0.75
PS C:\Users\binte\Desktop\sets>
```

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Section: B

Assignment: PF Lab week 12

Date: 25 March 2026

CLO2

Lab Tasks

Lambda Functions

Task 1:

task 2.py > ...

```python
find_max = lambda x, y: x if x > y else y


def table_print(num, range_limit):
    for i in range(1, range_limit + 1):
        print(num, "x", i, "=", num * i)

num1 = int(input("first number: "))
num2 = int(input("second number: "))
limit = int(input("Table kahan tak chahiye? "))

bada_num = find_max(num1, num2)
print("Bada number hai:", bada_num)
print("--- Table ---")
table_print(bada_num, limit)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
12 x 27 = 324
12 x 28 = 336
12 x 29 = 348
12 x 30 = 360
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 2:

```python
# Uppercase karne ke liye lambda
to_upper = lambda s: s.upper()

# String ko ulta karne ka basic UDF
def invert(text):
    # Python mein [::-1] string ko direct ulta kar deta hai
    print("Ulta string:", text[::-1])

user_string = input("String likhein: ")

uppercase_string = to_upper(user_string)
invert(uppercase_string)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/task 3.py"
String likhein: 30
Ulta string: 03
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 3:

```python
add = lambda a, b: a + b

print(add(10, 20))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pyth
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
30
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 4:

```python
square = lambda x: x * x

print(square(5))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Pytho
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
25
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 5:

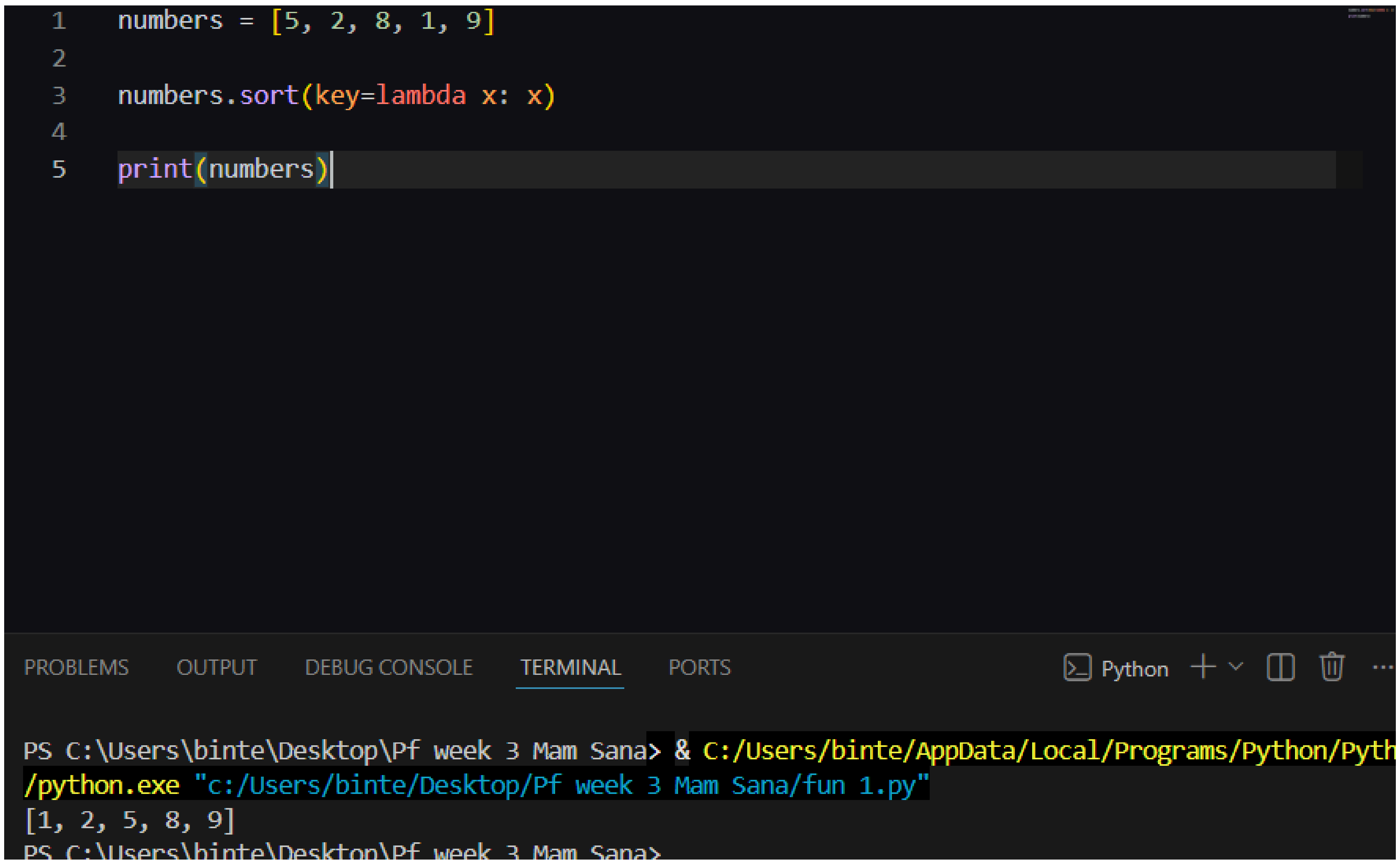
```python
even = lambda x: x % 2 == 0

print(even(8))
print(even(5))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python3
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
True
False
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 6:



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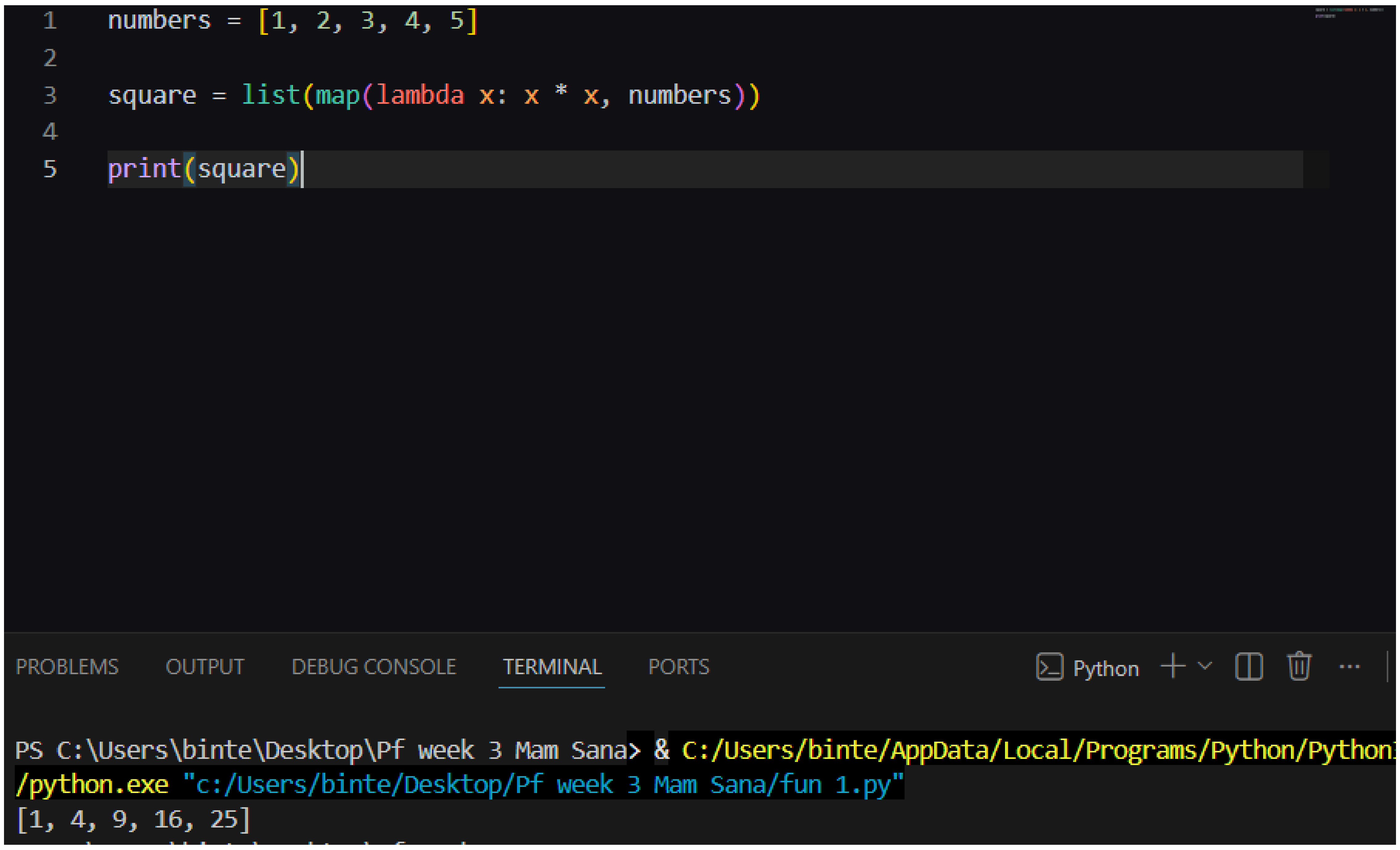
Date: 25 May 2026

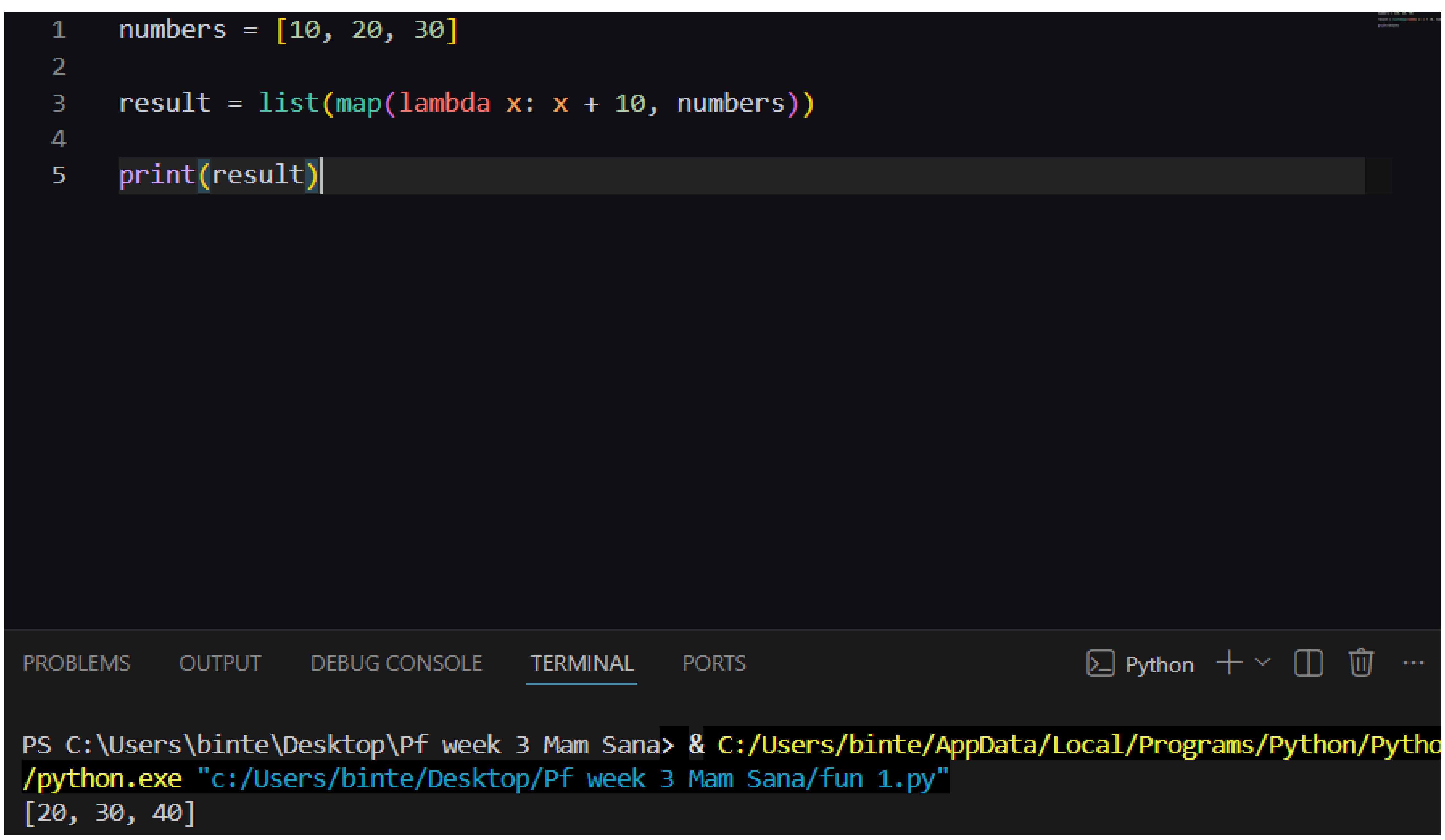
CLO2

Lab Tasks

Task 1:


numbers = [1, 2, 3, 4, 5]
square = list(map(lambda x: x * x, numbers))
print(square)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
[1, 4, 9, 16, 25]

Task 2:


numbers = [10, 20, 30]
result = list(map(lambda x: x + 10, numbers))
print(result)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pytho
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
[20, 30, 40]

Task 3:

```python
numbers = [1, 2, 3, 4, 5, 6]

even = list(filter(lambda x: x % 2 == 0, numbers))

print(even)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python3
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
[2, 4, 6]
```

Task 4:

```python
marks = [35, 60, 80, 45, 90]

passed = list(filter(lambda x: x >= 50, marks))

print(passed)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Python314
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
[60, 80, 90]
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```

Task 5:

```python
from functools import reduce

numbers = [1, 2, 3, 4, 5]

total = reduce(lambda x, y: x + y, numbers)

print(total)
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                Python

PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Progr
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
15
```

Task 6:

```python
from functools import reduce

numbers = [2, 3, 4]

result = reduce(lambda x, y: x * y, numbers)

print(result)
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS          Python

PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pyth
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
24
```

Task 7:

```python
names = ["ali", "sara", "ahmed"]

result = list(map(lambda x: x.upper(), names))

print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pytho
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
['ALI', 'SARA', 'AHMED']
```

Task 8:

```python
numbers = [10, 15, 20, 25, 30, 35]

odd = list(filter(lambda x: x % 2 != 0, numbers))

print(odd)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana> & C:/Users/binte/AppData/Local/Programs/Python/Pytho
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
[15, 25, 35]
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```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Assignment: PF Lab week 14

Date: 1 June 2026

CLO2

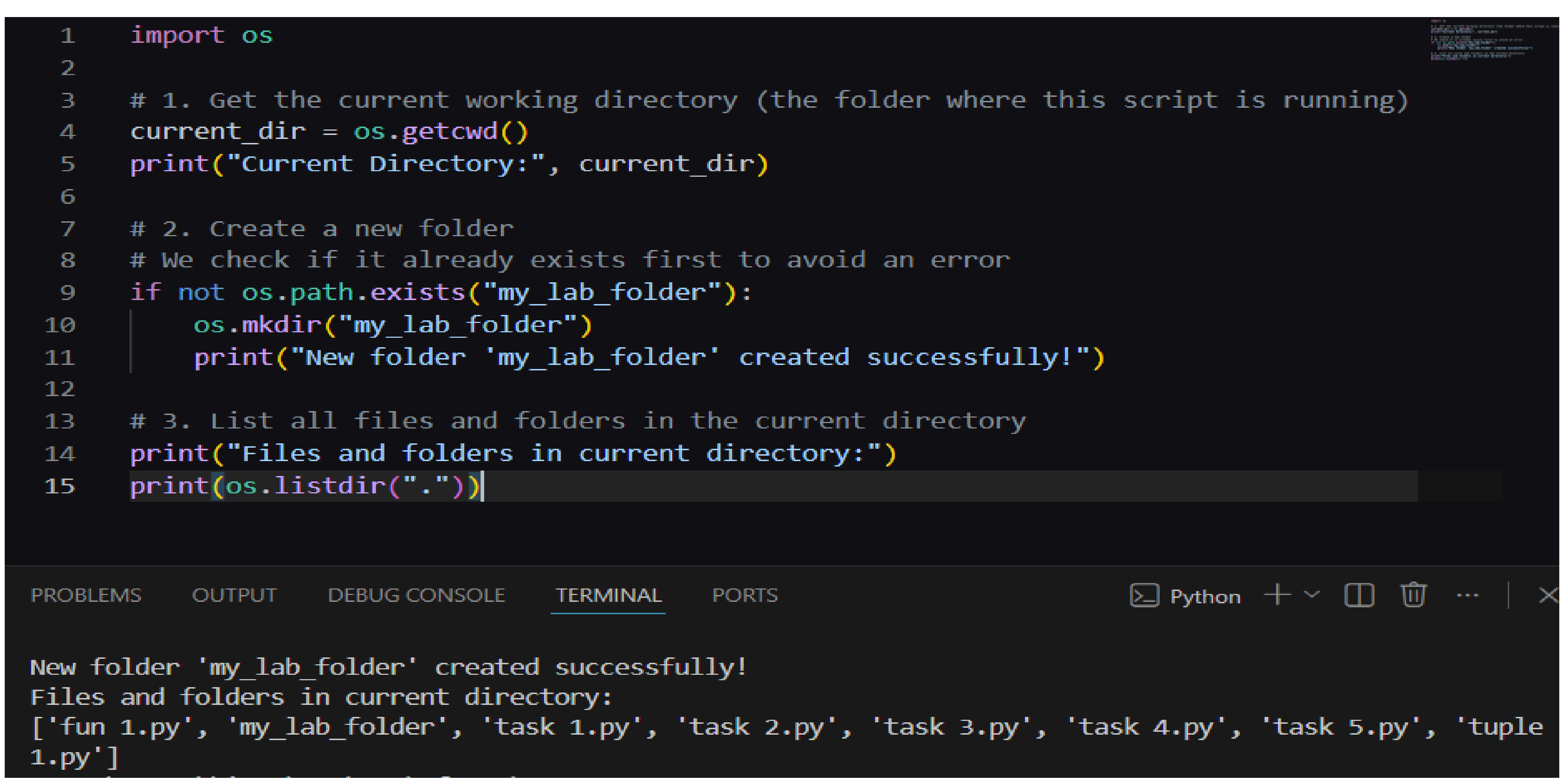
Lab Tasks

OS Library, if_name_==”_main_”, is vs ==

Task 1

1. OS Library

The os module in Python allows you to interact with your operating system (like creating folders, checking paths, and listing files).



```python
import os

# 1. Get the current working directory (the folder where this script is running)
current_dir = os.getcwd()
print("Current Directory:", current_dir)

# 2. Create a new folder
# We check if it already exists first to avoid an error
if not os.path.exists("my_lab_folder"):
    os.mkdir("my_lab_folder")
    print("New folder 'my_lab_folder' created successfully!")

# 3. List all files and folders in the current directory
print("Files and folders in current directory:")
print(os.listdir("."))
```

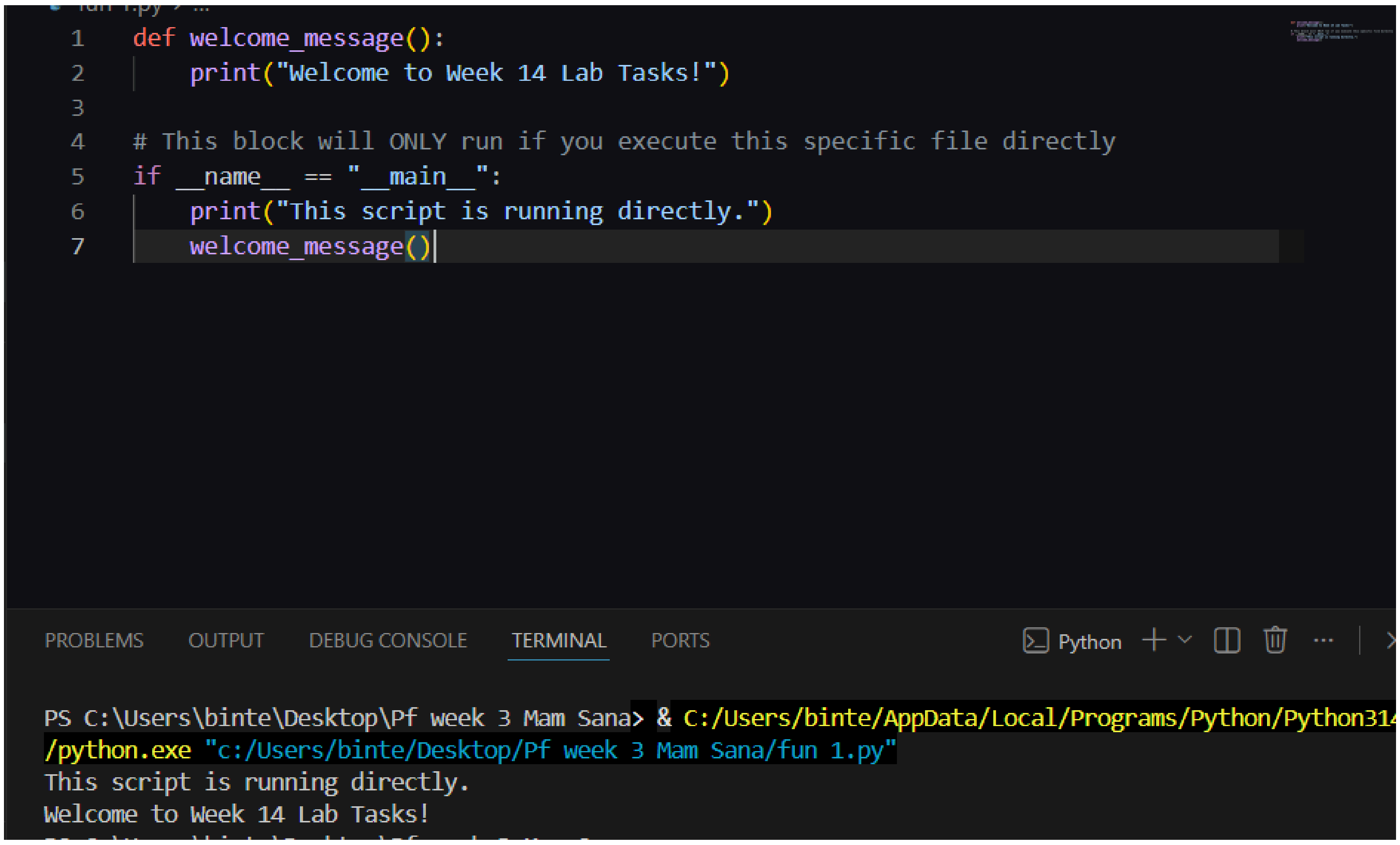


```
New folder 'my_lab_folder' created successfully!
Files and folders in current directory:
['fun 1.py', 'my_lab_folder', 'task 1.py', 'task 2.py', 'task 3.py', 'task 4.py', 'task 5.py', 'tuple 1.py']
```

Task 2:

2. if _name_ == "_main_":

This statement is used to determine whether the Python script is being *run directly* or being *imported* as a module into another script.



If you run this file directly, it will print the message. If you import this file into another Python file, the code inside the if block will be ignored until you explicitly call the function.

Task 3:

3. is vs == (Comparison Operators)

**** == checks if the *values* of two objects are equal.***

**** is checks if the two variables point to the *exact same object in memory* (Identity).***

```python
# Create two separate lists with identical values
list_a = [1, 2, 3]
list_b = [1, 2, 3]

# Point list_c to the exact same memory location as list_a
list_c = list_a

# 1. Using '==' to check equality of values
print("list_a == list_b:", list_a == list_b)  # True (Their content is identical)

# 2. Using 'is' to check object identity (Memory Location)
print("list_a is list_b:", list_a is list_b)  # False (They are two different objects in
print("list_a is list_c:", list_a is list_c)  # True (They point to the exact same memory
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
/python.exe "c:/Users/binte/Desktop/Pf week 3 Mam Sana/fun 1.py"
list_a == list_b: True
list_a is list_b: False
list_a is list_c: True
PS C:\Users\binte\Desktop\Pf week 3 Mam Sana>
```